Topic 5: Solved Problems

Step-by-step solutions of all 15 problems of the official sheet on mechanical properties (Problems T5_MechanicalProperties: the tensile test, the elastic modulus, yield and tensile strength, ductility, resilience, true stress and strain, thermal stress and hardness, from Sessions 11 and 12), written in four phases. The sheet has no worked solutions, only a numeric key for most items, so every solution is derived here and compared with that key.

`Problem 1` `Problems T5_MechanicalProperties.pdf #1` `Tensile test · full stress-strain analysis`

Complete Analysis of a Tensile Test from the Load-Elongation Table

`Original statement · Problems T5_MechanicalProperties.pdf`

1. A cylindrical aluminum sample has a diameter of 12.8 mm and an initial base length of 12.5 mm and it is subjected to uniaxial tension. Use the data from the load/elongation table to answer the following questions:

- *a. Represent the engineering stress data versus the engineering strain.*
- *b. Calculate Young's modulus.*
- *c. Calculate the yield strength, assuming that the residual strain at the yield point is 0.002.*
- *d. Work out the tensile strength of this alloy.*
- *e. Determine the approximate value of ductility expressed as a deformation percentage.*
- *f. Determine the approximate value of the modulus of resilience.*
- *g. Calculate the load (N) and the elongation (mm) of the sample at the yield point.*
- *h. Calculate an approximate working stress for this material.*
- *i. With the values obtained, is it possible to calculate the load applied to the sample if the elongation produced is 0.1 mm? If it is possible, work it out and if it is not possible, explain your answer and provide an alternative method.*

Point	F (N/100)	ΔL (mm)
1	127	0.006
2	254	0.012
3	381	0.019
4	508	0.025
5	762	0.038
6	891	0.051
7	927	0.063
8	1025	0.095
9	1078	0.127
10	1194	0.190

Point	F (N/100)	ΔL (mm)
11	1283	0.254
12	1497	0.508
13	1590	0.762
14	1604	0.889
15	1595	1.016
16	1515	1.270
17	1247	1.460

(The sheet prints the table in two rows, F and ΔL, followed by the word "Fracture"; here each point has its own row. Point 17 is the last record before the fracture.)

Official Solution: b. $E \approx 200$ **GPa; c.** $\sigma_y = 740$ **MPa; d.** $\sigma_{max} = 1250$ **MPa; e. 11.3 %; f.** $U_r = 1.37$ **MPa; g.** $F = 9.5 \times 10^4$ **N,** $\Delta l = 0.071$ **mm; h.** $\sigma_w = 370$ **MPa; i. it can only be obtained graphically,** $F = 10.3 \times 10^4$ **N**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The load acts along the axis of the cylinder, and up to the necking the stress is the same on every cross-section of the gauge length. Under this hypothesis the engineering stress and strain are defined with the original area and length [Session 11 Slide 9].
2. **Linear elastic start.** For small loads the curve is a straight line through the origin, Hooke's law $\sigma = E\varepsilon$, and the slope of that line is the Young's modulus [Session 11 Slide 13].
3. **Conventional yield strength.** The yield strength is the stress at which the curve crosses the line drawn parallel to the elastic line from the residual strain 0.002 (0.2 % offset), as the statement asks [Session 11 Slide 28].
4. **Between table rows the curve is a straight segment.** The table gives discrete points, so every reading between two of them is a linear interpolation.
5. **Elastic recovery.** When the sample breaks the load disappears and the elastic strain is recovered, so the ductility is the plastic strain at fracture [Session 11 Slide 32].
6. **Safety factor.** The statement gives none; the slide says that N is usually 2, and that value is used for the working stress [Session 11 Slide 29].

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	12.8 mm	12.8×10^{-3} m
Initial gauge length	l_0	12.5 mm	12.5×10^{-3} m
Load (table)	F	$100 \times (F/100)$	N
Elongation (table)	ΔL	see table	mm

Quantity	Symbol	Value	In SI
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{max} (d), the ductility (e), U_r (f), F_y and Δl_y (g), σ_w (h), and the load at $\Delta L = 0.1$ mm (i).

PHASE 2 · FRAMES & STRATEGY

There is a single direction, the axis of the sample, and the stress state is uniaxial, so no change of basis and no rotation matrix are involved.

Strategy. Every item comes from the same table of engineering stress and strain, so the first thing to do is to build it:

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

- The Young's modulus is the slope of the straight initial part (Hooke's law). A straight line through the origin is fitted by least squares: it minimises $\sum(\sigma_i - E\varepsilon_i)^2$, which gives $E = \sum \sigma_i\varepsilon_i / \sum \varepsilon_i^2$.
- The yield strength is found by the 0.2 % offset construction, solved algebraically on the segment of the table where the curve crosses the offset line.
- The tensile strength is the maximum nominal stress, and since A_0 is constant it corresponds to the maximum load.
- The ductility is the plastic strain at fracture, and the modulus of resilience is the area under the elastic line up to the yield point, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35].
- For the load at a given elongation the table is used again: Hooke's law only holds in the elastic zone.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(12.8\ \text{mm})^2}{4}$$

The square is $12.8^2 = 163.84$, then $\pi \times 163.84 = 514.72$ and dividing by 4:

$$A_0 = 128.68\ \text{mm}^2 = 1.2868 \times 10^{-4}\ \text{m}^2$$

2. Engineering stress and strain of every point

The load of the table is in hundreds of newtons, so $F = 100 \times (F/100)$. For point 1:

$$F_1 = 100 \times 127 = 12\,700\ \text{N} \qquad \sigma_{n,1} = \frac{12\,700\ \text{N}}{128.68\ \text{mm}^2} = 98.69\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{0.006\ \text{mm}}{12.5\ \text{mm}} = 4.80 \times 10^{-4}$$

The same operation for all the points (the last column is the secant modulus, used in item b):

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	12 700	0.006	98.69	0.00048	205.6
2	25 400	0.012	197.39	0.00096	205.6
3	38 100	0.019	296.08	0.00152	194.8
4	50 800	0.025	394.78	0.00200	197.4
5	76 200	0.038	592.17	0.00304	194.8
6	89 100	0.051	692.42	0.00408	169.7
7	92 700	0.063	720.39	0.00504	142.9
8	102 500	0.095	796.55	0.00760	104.8
9	107 800	0.127	837.74	0.01016	82.5
10	119 400	0.190	927.89	0.01520	61.0
11	128 300	0.254	997.05	0.02032	49.1
12	149 700	0.508	1163.35	0.04064	28.6
13	159 000	0.762	1235.63	0.06096	20.3
14	160 400	0.889	1246.51	0.07112	17.5
15	159 500	1.016	1239.51	0.08128	15.2
16	151 500	1.270	1177.34	0.10160	11.6
17	124 700	1.460	969.07	0.11680	8.3

a) Stress-strain curve

The points of the table are plotted as they are, joined by straight segments. The right-hand panel enlarges the elastic part, where the Young's modulus line and the offset line of item c are drawn.

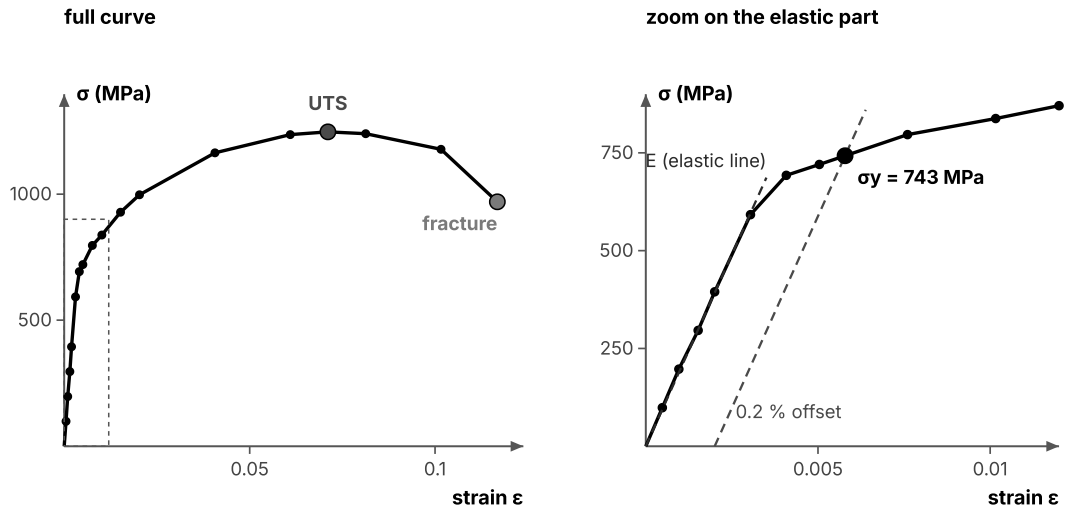


Figure 1. Problem 1: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Young's modulus

The secant modulus σ_n/ε_n of the last column is almost constant, 195 to 206 GPa, for points 1 to 5, and it drops to 170 GPa at point 6 and to 143 GPa at point 7: the specimen leaves the linear zone after point 5. The straight part is therefore made of points 1 to 5, and the fit through the origin is:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00048	98.69	0.0474	2.304×10^{-7}
2	0.00096	197.39	0.1895	9.216×10^{-7}
3	0.00152	296.08	0.4500	2.310×10^{-6}
4	0.00200	394.78	0.7896	4.000×10^{-6}
5	0.00304	592.17	1.8002	9.242×10^{-6}

$$\sum \sigma_i\varepsilon_i = 3.277 \text{ MPa} \qquad \sum \varepsilon_i^2 = 1.6704 \times 10^{-5}$$

$$E = \frac{\sum \sigma_i\varepsilon_i}{\sum \varepsilon_i^2} = \frac{3.277}{1.6704 \times 10^{-5}} = 196160 \text{ MPa} = \mathbf{196.2\ GPa} \approx 2.0 \times 10^{2} \text{ GPa}$$

c) Yield strength (0.2 % offset)

The offset line starts at $\varepsilon = 0.002$ and is parallel to the elastic line:

$$\sigma_{off}(\varepsilon) = E\,(\varepsilon - 0.002)$$

The curve is above the line while $g = \sigma_n - E(\varepsilon_n - 0.002)$ is positive. The sign of g changes between points 7 and 8, so the crossing is on that segment:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
6	0.00408	692.4	408.0	284.4
7	0.00504	720.4	596.3	124.1
8	0.00760	796.6	1098.5	-301.9
9	0.01016	837.7	1600.7	-762.9

The segment between the two points is the line

$$\sigma(\varepsilon) = \sigma_7 + m\,(\varepsilon - \varepsilon_7) \qquad m = \frac{\sigma_8 - \sigma_7}{\varepsilon_8 - \varepsilon_7} = \frac{796.55 - 720.39}{0.00760 - 0.00504} = 29749.3\ \text{MPa}$$

Equating it to the offset line, $\sigma_7 + m(\varepsilon - \varepsilon_7) = E(\varepsilon - 0.002)$, and solving for ε:

$$\varepsilon^{*} = \frac{\sigma_7 - m\,\varepsilon_7 + 0.002\,E}{E - m} = \frac{720.39 - 29749.3 \times 0.00504 + 0.002 \times 196160}{196160 - 29749.3} = \frac{962.778}{166411.2} = 0.0057855$$

$$\sigma_y = E\,(\varepsilon^{*} - 0.002) = 196160 \times (0.0057855 - 0.002) = \mathbf{742.6\ MPa}$$

d) Tensile strength

The maximum load of the table is $F_{max} = 160\,400$ N at point 14. The nominal stress uses the original area, so the maximum load gives the maximum stress:

$$\sigma_{max} = \frac{F_{max}}{A_0} = \frac{160\,400\ \text{N}}{128.68\ \text{mm}^2} = \mathbf{1246.5\ MPa}$$

e) Ductility

The last record before the fracture is point 17: $\Delta L = 1.460$ mm and $F = 124\,700$ N, so

$$\varepsilon_f = \frac{1.460}{12.5} = 0.1168 \qquad \sigma_f = \frac{124\,700}{128.68} = 969.1 \text{ MPa}$$

At that moment the strain is the sum of an elastic and a plastic part, $\varepsilon_t = \varepsilon_e + \varepsilon_p$. After the fracture the load is zero and the elastic part is recovered along a line parallel to the elastic one, so what remains is ε_p [Session 11 Slide 32]:

$$\varepsilon_e = \frac{\sigma_f}{E} = \frac{969.1}{196160} = 0.00494 \qquad \varepsilon_p = \varepsilon_f - \varepsilon_e = 0.1168 - 0.00494 = 0.1119$$

$$EL\,(\%) = \varepsilon_p \times 100 = \mathbf{11.2\ \%}$$

f) Modulus of resilience

The area under the elastic line up to the yield point is a triangle, $U_r = \tfrac{1}{2}\sigma_y \varepsilon_y$ with $\varepsilon_y = \sigma_y / E$ [Session 11 Slide 35]:

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(742.6 \text{ MPa})^2}{2 \times 196160 \text{ MPa}} = \mathbf{1.41\ MPa}$$

($1 \text{ MPa} = 1 \text{ MJ/m}^3$.)

g) Load and elongation at the yield point

The load is the yield stress times the original area, and the elongation is the total strain of the yield point (the value ε^* of item c) times the original length:

$$F_y = \sigma_y A_0 = 742.6 \text{ MPa} \times 128.68 \text{ mm}^2 = 95\,554 \text{ N} = \mathbf{95.6\ kN}$$

$$\Delta l_y = \varepsilon^* l_0 = 0.0057855 \times 12.5 \text{ mm} = \mathbf{0.0723\ mm}$$

h) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{742.6}{2} = \mathbf{371.3\ MPa}$$

i) Load for an elongation of 0.1 mm

The elongation of the yield point is 0.0723 mm, so 0.1 mm is in the plastic zone and the load is not proportional to the elongation. Using Hooke's law anyway, $F = EA_0\Delta l/l_0$, would give

$$F = \frac{196160\ \mathrm{N/mm^2} \times 128.68\ \mathrm{mm^2} \times 0.1\ \mathrm{mm}}{12.5\ \mathrm{mm}} = 201\,935\ \mathrm{N} = 202\ \mathrm{kN}$$

which is larger than the maximum load that the sample supports, 160 400 N: the answer is **not** possible with the values obtained. The alternative is the table itself. The elongation 0.1 mm is between points 8 ($\Delta L = 0.095$ mm, $F = 102\,500$ N) and 9 ($\Delta L = 0.127$ mm, $F = 107\,800$ N), and a linear interpolation on that segment gives

$$F = F_8 + \frac{0.1 - 0.095}{0.127 - 0.095}\,(F_9 - F_8) = 102\,500 + 0.15625 \times (107\,800 - 102\,500) = \mathbf{103\,328\ N}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\sigma_n = F/A_0$ in N/mm² is MPa, ε_n is a ratio of lengths and has no units, so $E = \sigma/\varepsilon$ is in MPa and $U_r = \sigma_y^2/2E$ is in MPa (energy per unit volume).
- **Order.** $\sigma_y = 742.6$ MPa $< \sigma_{max} = 1246.5$ MPa, and the fracture stress (969.1 MPa) is below the maximum, as in a ductile sample that necks.
- **The fit.** The secant moduli of points 1 to 5 are within 4.8 % of the fitted E, and the offset line crosses the curve between points 7 and 8, inside the segment used.
- **Item i.** The interpolated load, 103.3 kN, is between the loads of the two rows and below the maximum load; the Hooke value, 202 kN, is not.

Interpretation

The sample is stiff ($E \approx 196$ GPa) and strong, and it yields at about 742.6 MPa, a strain of 0.579 %. From there to the maximum load the strain hardening raises the stress by 504 MPa, and the 11.2 % of plastic elongation at fracture shows that it is also ductile.

THE DATA ARE NOT THOSE OF ALUMINIUM

The table of Session 11 gives $E = 69$ GPa for aluminium and 207 GPa for steel [Session 11 Slide 17]. The data of this problem give $E \approx 196$ GPa and a yield strength above 700 MPa, which are the values of a steel, not of aluminium. The statement says "aluminum"; the data are used as they are.

Check against the official solution

The key printed on the sheet is compared with the values computed here:

Item	Official	Computed	Comment
b. E	≈ 200 GPa	196.2 GPa	agrees (2 %)
c. σ_y	740 MPa	742.6 MPa	agrees
d. σ_{max}	1250 MPa	1246.5 MPa	agrees (rounded)
e. ductility	11.3 %	11.2 %	agrees within the reading
f. U_r	1.37 MPa	1.41 MPa	agrees (2.5 %); the key uses $E = 200$ GPa and $\sigma_y = 740$ MPa
g. F_y, Δl_y	9.5×10^4 N, 0.071 mm	95 554 N, 0.0723 mm	agrees
h. σ_w	370 MPa	371.3 MPa	agrees
i. load at 0.1 mm	10.3×10^4 N, graphically	103\,328 N	agrees

READING OF THE TABLE

The text of the PDF prints the first load of the table as "1127", but the figure of the sheet shows "127". With 127 the first point gives $\sigma/\varepsilon = 206$ GPa, in line with the rest of the elastic points and with the key, so 127 (12 700 N) is used.

`Problem 2` `Problems T5_MechanicalProperties.pdf #2` `Tensile test · true stress · area reduction`

Tensile Test of a 15 mm Cylinder: Strength, Resilience, True Stress and Area Reduction

`Original statement · Problems T5_MechanicalProperties.pdf`

2. A cylindrical sample made from a metallic material (assume a Poisson ratio of ν = 0.25) of 15 mm in diameter and a calibrated length of 50 mm was subjected to a tensile test, and the results are shown in the table below:

Point	Force (kN)	ΔL (mm)
1	70	0.25
2	120	0.40
3	150	0.50
4	160	0.60
5	170	0.75
6	200	1.75
7	220	3.00
8	233	5.00
9	233	6.50
10	220	8.00

The diameter of the sample after fracture was 12.45 mm. Calculate:

- *a. The tensile strength.*
- *b. The Young's Modulus (or elastic modulus).*
- *c. The yield stress.*
- *d. The modulus of resilience.*
- *e. The real tension for an engineering deformation of 0.02.*
- *f. The percentage of deformation to fracture and percentage of area reduction.*

(The sheet prints the table in two rows; here each point has its own row.)

Official Solution: a. $\sigma_{UTS} = 1319$ **MPa; b.** $E = 85$ **GPa; c.** $\sigma_y = 920$ **MPa; d.** $U_r = 4.98$ **MPa; e.** $\sigma_r = 1030$ **MPa; f. 15 %, 7.3 %**

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Uniaxial, uniform stress.** Up to the necking the stress is uniform along the gauge length, so the engineering quantities use the original area and length [Session 11 Slide 9].

2. **Linear elastic start.** Hooke's law $\sigma = E\varepsilon$ holds on the first points of the curve [Session 11 Slide 13].
3. **Yield strength by the 0.2 % offset.** The statement does not name a criterion; the conventional one for metals is the 0.2 % offset [Session 11 Slide 28].
4. **Straight segments between table rows**, so readings between two rows are linear interpolations.
5. **Uniform deformation before the necking.** The volume is constant, $A\,l = A_0 l_0$, until the maximum load, where the necking starts [Session 11 Slides 30 and 38]. The true stress of item e is computed in that range.
6. **Elastic recovery after the fracture.** The elastic strain disappears when the load does, and what is left is the plastic strain [Session 11 Slide 32]. The reduction of area uses the measured final diameter [Session 11 Slide 33].
7. **The Poisson ratio is not needed.** $\nu = 0.25$ is a datum of the statement, but none of the six items uses it: the only lateral information is the final diameter, which is measured.

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	15 mm	15×10^{-3} m
Gauge length	l_0	50 mm	50×10^{-3} m
Diameter after fracture	d_f	12.45 mm	12.45×10^{-3} m
Poisson ratio (not used)	ν	0.25	
Load, elongation	F, ΔL	table	kN, mm

Unknowns: σ_{UTS} (a), E (b), σ_y (c), U_r (d), σ_r at $\varepsilon_n = 0.02$ (e), the elongation at fracture and the reduction of area (f).

PHASE 2 · FRAMES & STRATEGY

The only direction is the axis of the cylinder and the stress is uniaxial, so there is no change of basis.

Strategy. As in any tensile test, the first step is to turn the table into engineering stress and strain,

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

and then read each item from it: the maximum stress (a), the slope of the straight part by least squares through the origin (b), the offset construction solved on the crossing segment (c), $U_r = \sigma_y^2 / 2E$ (d) [Session 11 Slide 35].

True stress (e). With constant volume, $A\,l = A_0 l_0$, so $A = A_0 l_0 / l = A_0/(1+\varepsilon_n)$. The true stress is then

$$\sigma_r = \frac{F}{A} = \frac{F}{A_0}\,(1+\varepsilon_n) = \sigma_n\,(1+\varepsilon_n)$$

which is the relation of the slide for the uniform deformation [Session 11 Slide 38].

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15\ \text{mm})^2}{4}$$

The square is $15^2 = 225$, then $\pi \times 225 = 706.86$ and dividing by 4:

$$A_0 = 176.71\ \text{mm}^2$$

2. Engineering stress and strain of every point

The loads are in kN, so $F = 1000 \times F_{kN}$ newtons. For example, point 1 is $F = 70\,000$ N and $\Delta L = 0.25$ mm, so $\sigma_n = 70\,000/176.71 = 396.1$ MPa and $\varepsilon_n = 0.25/50 = 0.0050$. All the points:

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	70 000	0.25	396.1	0.0050	79.2
2	120 000	0.40	679.1	0.0080	84.9
3	150 000	0.50	848.8	0.0100	84.9
4	160 000	0.60	905.4	0.0120	75.5
5	170 000	0.75	962.0	0.0150	64.1
6	200 000	1.75	1131.8	0.0350	32.3
7	220 000	3.00	1244.9	0.0600	20.7
8	233 000	5.00	1318.5	0.1000	13.2
9	233 000	6.50	1318.5	0.1300	10.1
10	220 000	8.00	1244.9	0.1600	7.8

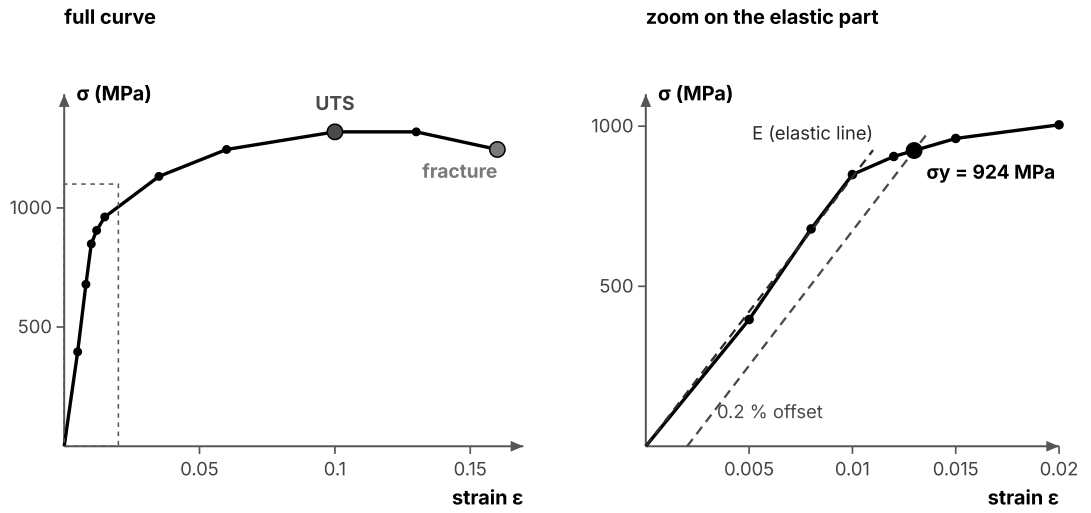


Figure 2. Problem 2: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

a) Tensile strength

The maximum load is $F_{max} = 233\,000$ N, reached at points 8 and 9 ($\varepsilon_n = 0.100$ and 0.130: the load stays at its maximum while the sample elongates). The area A_0 is constant in the engineering stress, so

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{233\,000\ \text{N}}{176.71\ \text{mm}^2} = \mathbf{1318.5\ MPa}$$

b) Young's modulus

The secant modulus is 79 to 85 GPa for points 1 to 3, and it drops to 75 GPa at point 4 and to 64 GPa at point 5: the linear zone ends after point 3. The fit through the origin with points 1 to 3 is:

Point	ε_i	σ_i (MPa)	$\sigma_i \varepsilon_i$ (MPa)	ε_i^2
1	0.0050	396.1	1.9806	2.500×10^{-5}
2	0.0080	679.1	5.4325	6.400×10^{-5}
3	0.0100	848.8	8.4883	1.000×10^{-4}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{15.901}{1.8900 \times 10^{-4}} = 84134\ \text{MPa} = \mathbf{84.1\ GPa}$$

c) Yield stress (0.2 % offset)

The offset line is $\sigma_{off} = E(\varepsilon - 0.002)$ and the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ tells where the curve crosses it. The sign changes between points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.01000	848.8	673.1	175.8
4	0.01200	905.4	841.3	64.1
5	0.01500	962.0	1093.7	-131.7
6	0.03500	1131.8	2776.4	-1644.7

On that segment

$$m = \frac{962.00 - 905.41}{0.01500 - 0.01200} = 18862.8 \text{ MPa} \qquad \varepsilon^* = \frac{\sigma_4 - m\,\varepsilon_4 + 0.002\,E}{E - m} = \frac{905.41 - 18862.8 \times 0.01200 + 0.002 \times 84134}{84134 - 18862.8} = \frac{847.329}{65271.3} = 0.0129817$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 84134 \times (0.0129817 - 0.002) = \mathbf{923.9\ MPa}$$

d) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(923.9\ \text{MPa})^2}{2 \times 84134\ \text{MPa}} = \mathbf{5.07\ MPa}$$

($1\ \text{MPa} = 1\ \text{MJ/m}^3$.)

e) True stress for $\varepsilon_n = 0.02$

The elongation that corresponds to $\varepsilon_n = 0.02$ is $\Delta L = 0.02 \times 50 = 1.00$ mm, which is not a row of the table. It is between points 5 ($\Delta L = 0.75$ mm, $F = 170$ kN) and 6 ($\Delta L = 1.75$ mm, $F = 200$ kN):

$$F = F_5 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(F_6 - F_5) = 170 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(200 - 170) = 177.5\ \text{kN}$$

$$\sigma_n = \frac{177\,500\ \text{N}}{176.71\ \text{mm}^2} = 1004.4\ \text{MPa}$$

This strain is below the one of the maximum load ($\varepsilon_n = 0.100$), so the deformation is still uniform and the relation of Phase 2 applies:

$$\sigma_r = \sigma_n\,(1+\varepsilon_n) = 1004.4 \times (1+0.02) = \mathbf{1024.5\ MPa}$$

f) Elongation at fracture and reduction of area

The last row of the table is the fracture: $\Delta L = 8.00$ mm and $F = 220\,000$ N, so

$$\varepsilon_f = \frac{8.00}{50} = 0.160 \qquad \sigma_f = \frac{220\,000}{176.71} = 1244.9\ \text{MPa}$$

The elastic part of that strain is recovered when the sample breaks, $\varepsilon_e = \sigma_f / E$, and what is left is the plastic strain [Session 11 Slide 32]:

$$\varepsilon_e = \frac{1244.9}{84134} = 0.01480 \qquad \varepsilon_p = 0.160 - 0.01480 = 0.1452 \qquad EL = \mathbf{14.5\ \%}$$

For the reduction of area, the final area from the final diameter is

$$A_f = \frac{\pi\,(12.45\ \text{mm})^2}{4} = \frac{\pi \times 155.0025}{4} = 121.74\ \text{mm}^2$$

$$RA\,(\%) = \frac{A_0 - A_f}{A_0} \times 100 = \frac{176.71 - 121.74}{176.71} \times 100 = \mathbf{31.1\ \%}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** Stresses in N/mm² = MPa, strains without units, E and U_r in MPa.
- **Order.** $\sigma_y = 923.9$ MPa $< \sigma_{UTS} = 1318.5$ MPa, and the stress rises 395 MPa between the yield and the maximum load.

- **True and engineering stress.** The true stress of item e, 1024.5 MPa, is 2 % above the engineering one, 1004.4 MPa, as the factor $1 + \varepsilon_n$ says.
- **Elongation against area reduction.** $RA = 31.1$ % is larger than $EL = 14.5$ %, as expected when a ductile sample necks: the section shrinks much more at the neck than the length grows over the whole gauge length.
- **Neck.** The stress that acts on the neck at the fracture is $F_f / A_f = 220\,000 \text{ N} / 121.74 \text{ mm}^2 = 1807$ MPa, far above the engineering fracture stress of 1244.9 MPa because the area is much smaller than A_0.

Interpretation

The sample reaches 1.3 GPa with a plateau of maximum load, and it fails after a clear neck. The measured final diameter, not Poisson's ratio, gives the lateral information: Poisson's ratio only describes the elastic part of the deformation.

Check against the official solution

Item	Official	Computed	Comment
a. σ_{UTS}	1319 MPa	1318.5 MPa	agrees
b. E	85 GPa	84.1 GPa	agrees (1 %); the key corresponds to the slope between points 2 and 3
c. σ_y	920 MPa	923.9 MPa	agrees (0.5 %)
d. U_r	4.98 MPa	5.07 MPa	agrees (2 %); the key uses $E = 85$ GPa and $\sigma_y = 920$ MPa
e. σ_r	1030 MPa	1024.5 MPa	agrees (0.5 %): the key reads $\sigma_n \approx 1010$ MPa at $\varepsilon_n = 0.02$
f. elongation	15 %	14.5 %	agrees (rounded)
f. area reduction	7.3 %	31.1 %	**does not agree**

DISCREPANCY WITH THE OFFICIAL SOLUTION: AREA REDUCTION

The official 7.3 % cannot be reproduced from the data of the statement. With $d_0 = 15$ mm and $d_f = 12.45$ mm the final area is $A_f = 121.74$ mm² and the reduction is 31.1 %. A reduction of 7.3 % would need a final area of $0.927\,A_0$, that is, a final diameter of $15\sqrt{0.927} = 14.44$ mm. The computed value is kept.

`Problem 3` `Problems T5_MechanicalProperties.pdf #3` `Tensile test · true stress and strain · design`

Tensile Test of a 13 mm Sample: Modulus, Cable Section and True Values at Maximum Load and Fracture

`Original statement · Problems T5_MechanicalProperties.pdf, p. 2`

3. An uniaxial tensile test was performed on a cylindrical sample of 13 mm in diameter getting the results of load and elongation shown in the table below:

- *a. Calculate the engineering stress and strain values and draw the stress-strain curve.*
- *b. Calculate the value of the elastic modulus (Young's Modulus) and determine the length of a bar of 13 cm on which a tension of 200 MPa is applied.*
- *c. What is the recommended cross-section for a cable that has to support an elevator with a maximum weight of 450 kg?*
- *d. Calculate the real stress and real strain values at maximum load and at the fracture point, assuming that the diameter of the sample is 12.922 mm and 10.348 mm respectively. Do the calculations without using the engineering stress and strain values.*
- *e. Calculate the modulus of resilience.*
- *f. Calculate the elastic strain at the moment before fracture.*

Point	Load (kg)	Calibrated length (mm)	Engineering stress (MPa)	Engineering strain
0	0	5.148		
1	500	5.151		
2	1500	5.156		
3	2500	5.161		
4	3500	5.166		
5	3750	5.225		
6	3950	5.354		
7	4000	5.457		
8	4000	5.560		
9	3600	5.676		

(In the sheet the table has only the first two columns filled; the last two are left blank to be completed in item a. Point 0 is the unloaded sample, points 7 and 8 have the same, maximum, load, and point 9 is the fracture.)

Official Solution: **b.** $E = 75.4$ **GPa,** $l = 13.04$ **cm; c.** $A_0 \geq 16.9$ **mm²; d. maximum load:** $\sigma_{UTS} = 300$ **MPa,** $\varepsilon_{UTS} = 0.012$**; fracture:** $\sigma_{fracture} = 420$ **MPa,** $\varepsilon_{fracture} = 0.456$**; e.** $U_r = 0.457$ **MPa; f.** $\varepsilon_y = 3.4 \times 10^{-3}$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress** up to the necking: engineering stress and strain with the original area and length [Session 11 Slide 9].
2. **Weight to force.** The loads are masses in kilograms; the force is $F = m\,g$ with $g = 9.81$ m/s².
3. **Linear elastic start** [Session 11 Slide 13] and **0.2 % offset yield strength** [Session 11 Slide 28].
4. **Calibrated length.** The first row ($m = 0$) gives the initial length $l_0 = 5.148$ mm and the other rows the current length. The strain is a ratio, so only the proportion matters.
5. **True quantities from the measured diameters** (item d). The true strain is $\varepsilon_r = \ln(A_0/A) = 2\ln(D_0/D)$ and the true stress is $\sigma_r = F/A$ with A the instantaneous area [Session 11 Slide 38]. They are valid also after the necking, because they use the actual area at the neck.
6. **Design stress (item c).** The statement does not give a safety factor. The minimum section is the one for which the cable does not yield, $\sigma = F/A_0 \le \sigma_y$; with the usual factor $N = 2$ it would be larger [Session 11 Slide 29].

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	13 mm	13×10^{-3} m
Initial calibrated length	l_0	5.148 mm	5.148×10^{-3} m
Gravity	g	9.81 m/s²	
Bar of item b: length, stress	l_b, σ	13 cm, 200 MPa	0.13 m, 200×10^6 Pa
Elevator (item c)	m	450 kg	
Diameter at maximum load, at fracture	D_m, D_f	12.922 mm, 10.348 mm	

Unknowns: the curve (a), E and the length of the bar (b), the section of the cable (c), the true stress and strain at maximum load and at fracture (d), U_r (e), the elastic strain before the fracture (f).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis) and uniaxial stress: there is no change of basis.

Strategy. The table is converted into $\sigma_n = F/A_0$ and $\varepsilon_n = (l - l_0)/l_0$. Then:

- E is the slope of the straight part (least squares through the origin), and the bar of item b is a direct application of Hooke's law, $\Delta l = (\sigma/E)\, l_b$.
- σ_y by the 0.2 % offset is needed twice: for the cable (c) and for the resilience (e).
- The true values (d) are computed from the forces and the measured diameters, as the statement asks, and not from the engineering values.
- The elastic strain before the fracture (f) is σ/E with the stress that acts at that moment [Session 11 Slide 32].

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section and conversions

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(13\ \text{mm})^2}{4}$$

With $13^2 = 169$ and $\pi \times 169 = 530.93$:

$$A_0 = 132.73\ \text{mm}^2$$

For the first loaded point, $F_1 = 500\ \text{kg} \times 9.81\ \text{m/s}^2 = 4\,905\ \text{N}$, so

$$\sigma_{n,1} = \frac{4\,905\ \text{N}}{132.73\ \text{mm}^2} = 36.95\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{5.151 - 5.148}{5.148} = 0.00058$$

a) Engineering stress and strain, and the curve

All the points:

Point	Load (kg)	l (mm)	$F = mg$ (N)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
0	0	5.148	0	0.00	0	—
1	500	5.151	4 905	36.95	0.00058	63.4
2	1500	5.156	14 715	110.86	0.00155	71.3

Point	Load (kg)	l (mm)	$F = mg$ (N)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
3	2500	5.161	24 525	184.77	0.00253	73.2
4	3500	5.166	34 335	258.68	0.00350	74.0
5	3750	5.225	36 788	277.16	0.01496	18.5
6	3950	5.354	38 750	291.94	0.04002	7.3
7	4000	5.457	39 240	295.63	0.06002	4.9
8	4000	5.560	39 240	295.63	0.08003	3.7
9	3600	5.676	35 316	266.07	0.10256	2.6

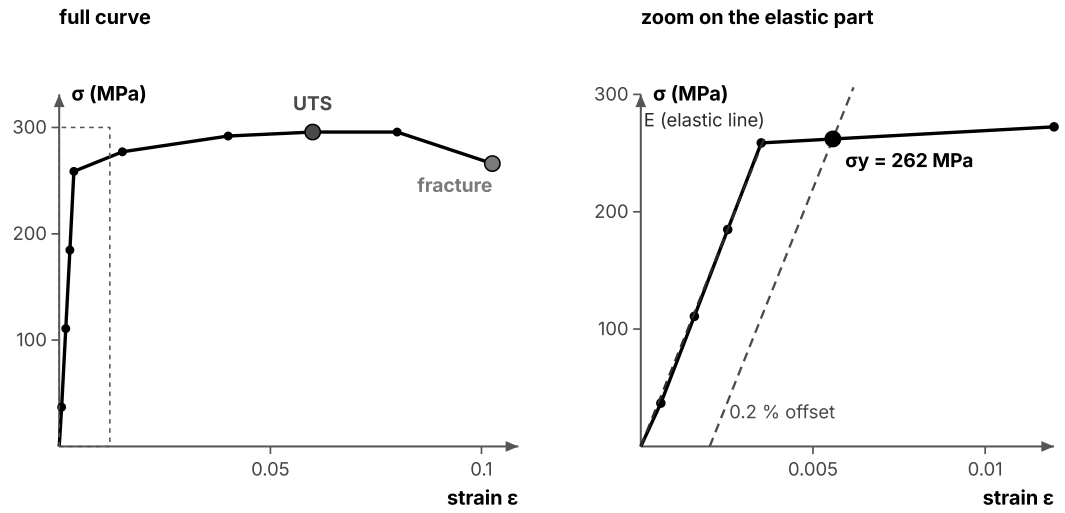


Figure 3. Problem 3: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Young's modulus and the length of the bar

The secant modulus of the last column grows from 63 to 74 GPa in points 1 to 4 and falls to 18 GPa at point 5: the linear zone ends after point 4. The first value is lower because a strain of 0.0006 comes from an elongation of only 0.003 mm, at the limit of the resolution of the table. The fit through the origin on points 1 to 4 is:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00058	36.95	0.0215	3.396×10^{-7}
2	0.00155	110.86	0.1723	2.415×10^{-6}
3	0.00253	184.77	0.4666	6.377×10^{-6}
4	0.00350	258.68	0.9045	1.223×10^{-5}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{1.565}{2.1357 \times 10^{-5}} = 73273 \text{ MPa} = \mathbf{73.3\ GPa}$$

For the bar, the strain produced by $\sigma = 200$ MPa is $\varepsilon = \sigma / E$ and the elongation is $\Delta l = \varepsilon\, l_b$:

$$\varepsilon = \frac{200}{73273} = 0.002730 \qquad \Delta l = 0.002730 \times 130 \text{ mm} = 0.355 \text{ mm}$$

$$l = l_b + \Delta l = 130 + 0.355 = 130.355 \text{ mm} = \mathbf{13.04\ cm}$$

c) Section of the cable

The yield strength is needed first. With the offset line $E(\varepsilon - 0.002)$ the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ changes between points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.00253	184.8	38.5	146.3
4	0.00350	258.7	109.7	149.0
5	0.01496	277.2	949.4	-672.3
6	0.04002	291.9	2785.5	-2493.6

$$m = \frac{277.16 - 258.68}{0.01496 - 0.00350} = 1612.2 \text{ MPa} \qquad \varepsilon^* = \frac{258.68 - 1612.2 \times 0.00350 + 0.002 \times 73273}{73273 - 1612.2} = \frac{399.587}{71660.3} = 0.0055761$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 73273 \times (0.0055761 - 0.002) = \mathbf{262.0\ MPa}$$

The weight of the elevator is $W = 450 \times 9.81 = 4414.5$ N. For the cable not to yield, $W / A_0 \le \sigma_y$:

$$A_0 \geq \frac{W}{\sigma_y} = \frac{4414.5\ \text{N}}{262.0\ \text{N/mm}^2} = \mathbf{16.85\ mm^2}$$

If the usual safety factor $N = 2$ is also applied, the design stress is $\sigma_y/2$ and the section doubles, $A_0 \geq 33.7$ mm².

d) True stress and strain at maximum load and at fracture

Maximum load. $F_m = 4000 \times 9.81 = 39\,240$ N and $D_m = 12.922$ mm:

$$A_m = \frac{\pi\,(12.922)^2}{4} = \frac{\pi \times 166.978}{4} = 131.14\ \text{mm}^2$$

$$\sigma_{r,UTS} = \frac{F_m}{A_m} = \frac{39\,240}{131.14} = \mathbf{299.2\ MPa} \qquad \varepsilon_{r,UTS} = 2\ln\frac{D_0}{D_m} = 2\ln\frac{13}{12.922} = 2\ln(1.00604) = 2 \times 0.006018 = \mathbf{0.01204}$$

Fracture. $F_f = 3600 \times 9.81 = 35\,316$ N and $D_f = 10.348$ mm:

$$A_f = \frac{\pi\,(10.348)^2}{4} = \frac{\pi \times 107.081}{4} = 84.10\ \text{mm}^2$$

$$\sigma_{r,f} = \frac{F_f}{A_f} = \frac{35\,316}{84.10} = \mathbf{419.9\ MPa} \qquad \varepsilon_{r,f} = 2\ln\frac{13}{10.348} = 2\ln(1.25628) = 2 \times 0.22816 = \mathbf{0.4563}$$

e) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(262.0\ \text{MPa})^2}{2 \times 73273\ \text{MPa}} = \mathbf{0.469\ MPa}$$

f) Elastic strain before the fracture

The stress of the table at the fracture point (point 9) is $\sigma_n = 266.1$ MPa, and the elastic part of the strain is given by Hooke's law:

$$\varepsilon_e = \frac{\sigma_n}{E} = \frac{266.1}{73273} = \mathbf{3.63 \times 10^{-3}}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** Masses in kg and g in m/s² give newtons, N/mm² is MPa, the strains have no units, and $A_0 \geq W/\sigma_y$ comes out in mm².
- **Order.** The true stress at maximum load (299.2 MPa) is above the engineering maximum (295.6 MPa) because the area has shrunk, and the true fracture stress (419.9 MPa) is much above the engineering one (266.1 MPa), while the engineering stress falls after the maximum.
- **Limit.** For $D = D_0$ the true strain $2\ln(D_0/D)$ is zero and the true stress is F/A_0, the engineering one.
- **Bar.** The elongation, 0.355 mm over 130 mm, is a strain of 0.27 %, well inside the elastic zone (the yield strain is 3.58×10^{-3} = 0.36 %).

Interpretation

The sample is a ductile metal with a clear yield at 262.0 MPa and a long hardening plateau up to 295.6 MPa; the neck makes the true stress at fracture 419.9 MPa. The cable needs at least 16.85 mm² to stay elastic under the elevator, and about twice as much with the usual safety factor.

DIAMETERS AND LENGTHS OF THE TABLE DO NOT DESCRIBE THE SAME STRAIN

At the maximum load the length of the table gives $\varepsilon_n = 0.0600$, that is, $\varepsilon_r = \ln(1+\varepsilon_n) = 0.0583$ with the uniform-deformation relation, and constant volume would give a diameter $13/\sqrt{1+\varepsilon_n} = 12.63$ mm. The statement gives 12.922 mm, with $\varepsilon_r = 0.01204$. At the fracture, $A_0/A_f = 1.578$ while $l/l_0 = 1.1026$. The two sets of data do not describe the same deformation: the diameters measure the strain of the neck, the lengths the average over the calibrated length. The statement asks to compute the true values from the diameters, and that is what is done.

Check against the official solution

Item	Official	Computed	Comment
b. E	75.4 GPa	73.3 GPa	agrees (3 %)
b. length of the bar	13.04 cm	13.04 cm	agrees
c. A_0	≥ 16.9 mm²	≥ 16.85 mm²	agrees: the key uses σ_y without safety factor
d. $\sigma_{r,UTS}$, $\varepsilon_{r,UTS}$	300 MPa, 0.012	299.2 MPa, 0.01204	agrees

Item	Official	Computed	Comment
d. $\sigma_{r,f}$, $\varepsilon_{r,f}$	420 MPa, 0.456	419.9 MPa, 0.4563	agrees
e. U_r	0.457 MPa	0.469 MPa	agrees (2.5 %): the key uses $E = 75.4$ GPa
f. elastic strain	3.4×10^{-3}	3.63×10^{-3}	differs 7 %

DISCREPANCY WITH THE OFFICIAL SOLUTION: ELASTIC STRAIN BEFORE THE FRACTURE

The key gives 3.4×10^{-3}. With the stress of the table at the fracture, 266.1 MPa, and the fitted modulus the elastic strain is 3.63×10^{-3}; with the key's modulus, 75.4 GPa, it is 3.53×10^{-3}. The yield strain, $\sigma_y / E = 3.58 \times 10^{-3}$, is also not the key value. The value cannot be reproduced exactly, and the computed one is kept.

Problem 4 | Problems T5_MechanicalProperties.pdf #4 | Tensile test · rectangular brass sample

Tensile Test of a Rectangular Brass Sample: Modulus, Strengths, Resilience and Working Stress

Original statement · Problems T5_MechanicalProperties.pdf, p. 2

4. A tensile test was performed on a brass sample with a rectangular section and dimensions of 3.2 mm x 19.1 mm. Using the data in the table below, answer the following questions:

- *a. Represent the data in terms of engineering stress versus strain.*
- *b. Calculate the modulus of elasticity (Young's modulus).*
- *c. Determine the yield strength.*
- *d. Determine the tensile strength.*
- *e. Calculate the modulus of resilience.*
- *f. Calculate the working stress of this material.*

Point	Load (N)	Length (mm)	σn (MPa)	εn
0	0	63.50		
1	1 380	63.53		
2	2 780	63.58		
3	5 630	63.63		
4	7 430	63.70		
5	8 140	63.75		
6	9 870	64.14		
7	12 850	65.41		
8	14 100	66.68		
9	14 300	67.95		
10	13 830	69.22		
11	12 500	70.49		

(The sheet prints the table in two rows, load and length, with two more rows, σn and εn, left blank to be completed. Point 0 is the unloaded sample.)

Official Solution: **b.** $E \approx 44$ **GPa;** **c.** $\sigma_y = 150$ **MPa;** **d.** $\sigma_{UTS} = 234$ **MPa;** **e.** $U_r = 0.28$ **MPa;** **f.** $\sigma_w = 75$ **MPa**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The section is a rectangle 3.2×19.1 mm, and up to the necking the stress is uniform: $\sigma_n = F/A_0$ and $\varepsilon_n = (l - l_0)/l_0$ [Session 11 Slide 9].
2. **Initial length.** The first value of the length row, 63.50 mm (load 0), is l_0.
3. **Linear elastic start**, Hooke's law, with the slope fitted by least squares through the origin [Session 11 Slide 13].
4. **0.2 % offset yield strength** [Session 11 Slide 28].
5. **Working stress** $\sigma_w = \sigma_y/N$ with the usual $N = 2$, which the statement does not give [Session 11 Slide 29].
6. **Resolution of the length.** The length is given to 0.01 mm, so the first elongations (0.03 and 0.08 mm) carry an error of 10 to 30 % of their value. This matters for the modulus, and it is analysed in item b.

Data

Quantity	Symbol	Value	In SI
Section	$b \times h$	19.1×3.2 mm	$19.1 \times 10^{-3} \times 3.2 \times 10^{-3}$ m
Initial length	l_0	63.50 mm	63.50×10^{-3} m
Load, length	F, l	table	N, mm
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{UTS} (d), U_r (e), σ_w (f).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), uniaxial stress, no change of basis.

Strategy. The table is turned into σ_n and ε_n. Then E is the least-squares slope through the origin of the straight part, σ_y comes from the offset construction on the segment that crosses the offset line, σ_{UTS} is the maximum load over A_0, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35] and $\sigma_w = \sigma_y/N$. Since the first points of this table are not perfectly straight, the result is also shown for different choices of points, to see how much it depends on the choice.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = b\,h = 19.1 \text{ mm} \times 3.2 \text{ mm} = 61.12 \text{ mm}^2$$

For point 1: $F_1 = 1\,380$ N and $l_1 = 63.53$ mm, so

$$\sigma_{n,1} = \frac{1\,380\ \text{N}}{61.12\ \text{mm}^2} = 22.58\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{63.53 - 63.50}{63.50} = 0.00047$$

a) Engineering stress and strain, and the curve

All the points:

Point	F (N)	l (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
0	0	63.50	0	0.00000	—
1	1 380	63.53	22.58	0.00047	47.8
2	2 780	63.58	45.48	0.00126	36.1
3	5 630	63.63	92.11	0.00205	45.0
4	7 430	63.70	121.56	0.00315	38.6
5	8 140	63.75	133.18	0.00394	33.8
6	9 870	64.14	161.49	0.01008	16.0
7	12 850	65.41	210.24	0.03008	7.0
8	14 100	66.68	230.69	0.05008	4.6
9	14 300	67.95	233.97	0.07008	3.3
10	13 830	69.22	226.28	0.09008	2.5
11	12 500	70.49	204.52	0.11008	1.9

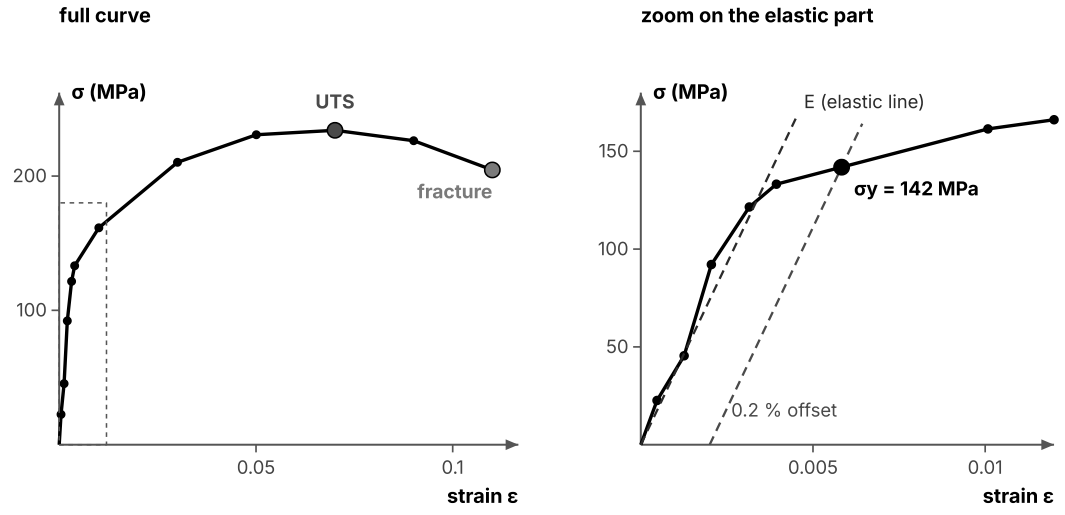


Figure 4. Problem 4: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Modulus of elasticity

The secant modulus of the last column is 34 to 48 GPa for points 1 to 5 and falls to 16 GPa at point 6: the straight part ends after point 5. The scatter between 34 and 48 GPa comes from the resolution of the length: point 1 is an elongation of 0.03 mm, and an error of 0.01 mm is 33 % of it. A fit through the origin on points 1 to 5 weights the points by their strain, so the points with larger elongation, and smaller relative error, count more:

Point	ε_i	σ_i (MPa)	$\sigma_i \varepsilon_i$ (MPa)	ε_i^2
1	0.00047	22.58	0.0107	2.232×10^{-7}
2	0.00126	45.48	0.0573	1.587×10^{-6}
3	0.00205	92.11	0.1886	4.191×10^{-6}
4	0.00315	121.56	0.3829	9.920×10^{-6}
5	0.00394	133.18	0.5243	1.550×10^{-5}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{1.164}{3.1422 \times 10^{-5}} = 37037 \text{ MPa} = \mathbf{37.0 \text{ GPa}}$$

c) Yield strength (0.2 % offset)

With the offset line $E(\varepsilon - 0.002)$ the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ changes between points 5 and 6:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
4	0.00315	121.6	42.6	79.0
5	0.00394	133.2	71.7	61.4
6	0.01008	161.5	299.2	-137.7
7	0.03008	210.2	1040.0	-829.7

$$m = \frac{161.49 - 133.18}{0.01008 - 0.00394} = 4608.6 \text{ MPa} \qquad \varepsilon^* = \frac{133.18 - 4608.6 \times 0.00394 + 0.002 \times 37037}{37037 - 4608.6} = \frac{189.110}{32428.3} = 0.0058316$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 37037 \times (0.0058316 - 0.002) = \mathbf{141.9 \text{ MPa}}$$

d) Tensile strength

The maximum load is $F_{max} = 14\,300$ N (point 9):

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{14\,300\ \text{N}}{61.12\ \text{mm}^2} = \mathbf{234.0\ MPa}$$

e) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(141.9\ \text{MPa})^2}{2 \times 37037\ \text{MPa}} = \mathbf{0.272\ MPa}$$

f) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{141.9}{2} = \mathbf{71.0\ MPa}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** N/mm² is MPa, strains have no units, E and U_r are in MPa.
- **Order.** $\sigma_y = 141.9$ MPa $< \sigma_{UTS} = 234.0$ MPa; after the maximum the load falls (point 11 has 12 500 N), the sample necks.
- **Sensitivity to the points chosen for E.** The modulus, and through it σ_y and U_r, depend on how many of the first points are taken:

Points used for E	E (GPa)	σ_y (MPa)	U_r (MPa)	σ_w (MPa)
1 to 3	42.7	139.3	0.227	69.6
1 to 4	40.2	140.4	0.245	70.2
1 to 5	37.0	141.9	0.272	71.0

Between the three choices the yield strength changes by 1.9 % and U_r by 20 % (because it contains $1/E$): the strengths hardly depend on the choice made for E, but the resilience does.

Interpretation

The brass has a tensile strength of 234.0 MPa and a yield strength of about 141.9 MPa, and between the two the strain hardening adds 92 MPa. The large strain at fracture (11 % at the last point, after the maximum load) shows that it is ductile.

THE MODULUS IS MUCH LOWER THAN THE ONE OF BRASS IN THE SLIDE

The table of Session 11 gives $E = 97$ GPa for brass [Session 11 Slide 17]. The data of this problem give 37 to 43 GPa, less than half. The key also gives 44 GPa, so the sheet itself uses these data. The reason is probably that the elongation of the table includes the slack and the compliance of the testing machine in the first points, which is why the elastic part looks softer than the real material. The data are used as they are.

Check against the official solution

Item	Official	Computed	Comment
b. E	≈ 44 GPa	37.0 GPa	differs 16 %: see the sensitivity table; with points 1 to 3 the fit gives 42.7 GPa
c. σ_y	150 MPa	141.9 MPa	differs 5 %
d. σ_{UTS}	234 MPa	234.0 MPa	agrees
e. U_r	0.28 MPa	0.272 MPa	agrees (3 %)
f. σ_w	75 MPa	71.0 MPa	consistent with σ_y

DISCREPANCY WITH THE OFFICIAL SOLUTION: MODULUS AND YIELD STRENGTH

The key gives $E \approx 44$ GPa and $\sigma_y = 150$ MPa. The 0.2 % offset on the data of the table gives σ_y between 139 MPa ($E = 44$ GPa) and 142 MPa ($E = 37.0$ GPa), never 150 MPa. A yield strength of 150 MPa corresponds to the stress where the curve bends clearly (points 5 and 6, 133 to 161 MPa), that is, to a reading by eye of the end of the straight part. In addition, with the key values, $U_r = 150^2/(2 \times 44\,000) = 0.256$ MPa, not 0.28 MPa; the computed resilience, 0.272 MPa, is close to the printed one only by coincidence. The computed values are kept.

`Problem 5` `Problems T5_MechanicalProperties.pdf #5` `Thermal stress · constant length`

Thermal Stress in a Constrained Steel Wire: the Diameter that Cancels the Applied Stress

`Original statement · Problems T5_MechanicalProperties.pdf, p. 3`

> *5. A steel wire, initially 1 meter in length, is stretched by a force of 1400 N at a temperature of 20 ºC. After the force is applied, the wire is held at a constant length and then heated to 40 ºC. Determine the required diameter of the wire to ensure that the thermal expansion stress compensates for the applied stress.*
>
> *Data: E = 207 GPa; $\alpha_l = 12.5 \times 10^{-6}$ ºC⁻¹;*

`Official Solution: 5.86 mm`

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Uniaxial, linear elastic wire.** The wire is loaded only along its axis and it stays elastic, so $\sigma = E\varepsilon$ [Session 11 Slide 13].
2. **Constant length while heating.** After the force is applied, the wire is held at a constant length: the total length, and therefore the total strain ε, does not change during the heating.
3. **Linear thermal expansion.** A temperature rise ΔT produces a free thermal strain $\varepsilon_T = \alpha_l \, \Delta T$. This law is not in the slides of Session 11 and 12 (it belongs to the thermal properties of Topic 7); it is given by the sheet through the datum α_l.
4. **Strains add.** The total strain is the elastic (mechanical) strain plus the thermal strain, $\varepsilon = \sigma/E + \alpha_l \, \Delta T$, and E and α_l are constant between 20 and 40 °C.
5. **"Compensates".** The thermal stress compensates the applied stress when the net stress of the wire after heating is zero.
6. **Circular, uniform section.** The area $A = \pi d^2/4$ is the same all along the wire and does not change with the temperature.

Data

Quantity	Symbol	Value	In SI
Force	F	1400 N	1400 N
Initial temperature, final temperature	T_1, T_2	20 °C, 40 °C	
Temperature rise	ΔT	20 °C	20 K
Young's modulus	E	207 GPa	207×10^9 Pa = 207 000 N/mm²

Quantity	Symbol	Value	In SI
Linear expansion coefficient	α_l	12.5×10^{-6} °C^{-1}	12.5×10^{-6} K^{-1}
Initial length	l_0	1 m	not needed

Unknown: the diameter d of the wire.

PHASE 2 · FRAMES & STRATEGY

There is a single direction, the axis of the wire; no change of basis is involved.

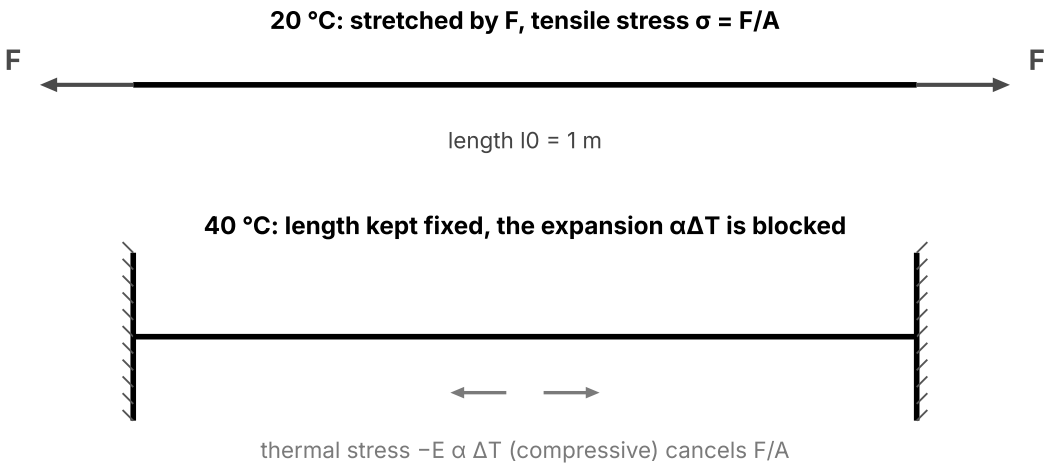


Figure 5. Problem 5: the wire is stretched at 20 °C and then held at constant length while it is heated to 40 °C (schematic).

Strategy. The key is to follow the strain, because the length is the quantity that is fixed.

1. At 20 °C the force produces the stress $\sigma_F = F/A$ and the strain $\varepsilon_1 = \sigma_F/E$.
2. At 40 °C the strain is the same, $\varepsilon_2 = \varepsilon_1$, but it is now the sum of an elastic part σ_2/E and a thermal part $\alpha_l \Delta T$. This gives the stress after the heating, σ_2.
3. The condition of the statement, $\sigma_2 = 0$, gives the area, and the area gives the diameter.

This is better than thinking in terms of forces, because the wire does not "know" about the force: it only keeps its length, and the stress is whatever is needed to keep it.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Stress after heating at constant length

Before heating, the strain is $\varepsilon_1 = \sigma_F / E$ with $\sigma_F = F/A$. After heating, the strain is the same, and it is the sum of the elastic and the thermal strain:

$$\varepsilon_2 = \frac{\sigma_2}{E} + \alpha_l \, \Delta T = \varepsilon_1 = \frac{\sigma_F}{E}$$

Solving for the stress after the heating:

$$\sigma_2 = \sigma_F - E \, \alpha_l \, \Delta T$$

The second term is the thermal stress: heating at constant length lowers the tension by $E \alpha_l \Delta T$ (if it were larger than σ_F, σ_2 would be negative, a compression).

2. Condition of the statement

The thermal stress compensates the applied stress when the net stress vanishes, $\sigma_2 = 0$:

$$\frac{F}{A} = E \, \alpha_l \, \Delta T \quad \Rightarrow \quad A = \frac{F}{E \, \alpha_l \, \Delta T}$$

3. Numerical value of the thermal stress

With $E = 207000$ N/mm²:

$$E \, \alpha_l = 207000 \, \frac{\mathrm{N}}{\mathrm{mm}^2} \times 12.5 \times 10^{-6} \, {}^\circ\mathrm{C}^{-1} = 2.5875 \, \frac{\mathrm{N}}{\mathrm{mm}^2 \, {}^\circ\mathrm{C}}$$

$$E \, \alpha_l \, \Delta T = 2.5875 \times 20 \, {}^\circ\mathrm{C} = 51.75 \, \mathrm{N/mm}^2 = 51.75 \, \mathrm{MPa}$$

4. Area and diameter

$$A = \frac{F}{E\alpha_l \Delta T} = \frac{1400\ \text{N}}{51.75\ \text{N/mm}^2} = 27.053\ \text{mm}^2$$

From $A = \pi d^2/4$, $d = \sqrt{4A/\pi}$:

$$4A = 108.21\ \text{mm}^2 \qquad \frac{4A}{\pi} = 34.445\ \text{mm}^2 \qquad d = \sqrt{34.445\ \text{mm}^2} = \mathbf{5.869\ mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $F/(E\alpha_l \Delta T)$: N divided by (N/mm² · °C⁻¹ · °C) = N/(N/mm²) = mm². The product $\alpha_l \Delta T$ has no units, so $E\alpha_l \Delta T$ is a stress.
- **Direct check.** With this area the applied stress is $F/A = 1400/27.053 = 51.75$ MPa, equal to the thermal stress 51.75 MPa, so $\sigma_2 = 0$.
- **Strains.** The thermal strain $\alpha_l \Delta T = 2.500 \times 10^{-4}$ is the same as the elastic strain that the force produced, $\sigma_F/E = 2.500 \times 10^{-4}$: heating gives back exactly the stretch that the force had imposed, and since the length cannot change the stress disappears.
- **Length not needed.** The initial length does not appear in the result. The free expansion of 1 m of wire would be $\alpha_l \Delta T\, l_0 = 0.25$ mm, the amount that the supports prevent.

Interpretation

For a wire thinner than 5.87 mm the applied stress is larger than the thermal one, and some tension remains after the heating. For a thicker wire the applied stress is smaller, and the heating leaves the wire in compression (it could buckle). The result depends on F/A, not on the length, because only the strain matters. As $\Delta T \to 0$ the required area tends to infinity: without temperature rise the force cannot be compensated.

Check against the official solution

Item	Official	Computed	Comment
Diameter	5.86 mm	5.869 mm	agrees (0.15 %): the key is truncated, 5.869 rounds to 5.87

Problem 6 | Problems T5_MechanicalProperties.pdf #6 | Hooke's law · design

Maximum Length of a Ni Alloy Sample for a Given Allowed Elongation

Original statement · Problems T5_MechanicalProperties.pdf, p. 3

6. A sample made of a Ni alloy with an elastic modulus of 207 GPa and an initial diameter of 10.2 mm experiences an elastic deformation only when loads inferior to 8900 N are applied. Calculate the maximum length of the sample before the deformation if the maximum elongation allowed is 0.25 mm.

Official Solution: $l_0 = 475$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Elastic up to 8900 N.** For loads below 8900 N the deformation is only elastic, so Hooke's law $\sigma = E\varepsilon$ holds [Session 11 Slide 13]. The load of 8900 N is therefore the largest load for which the sample is still elastic.
2. **Uniaxial, uniform stress** along the sample: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
3. **Worst case.** The elongation grows with the load and with the length. The longest sample that does not exceed the allowed elongation is found for the largest load that is applied, 8900 N.
4. The cross-section is a circle of 10.2 mm, and it does not change appreciably in the elastic range.

Data

Quantity	Symbol	Value	In SI
Young's modulus	E	207 GPa	207 000 N/mm²
Diameter	d_0	10.2 mm	10.2×10^{-3} m
Largest elastic load	F	8900 N	8900 N
Maximum elongation allowed	Δl_{max}	0.25 mm	0.25×10^{-3} m

Unknown: the maximum initial length l_0.

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), no change of basis.

Strategy. Hooke's law relates the stress produced by the load to the strain, and the strain relates the elongation to the length. Combining the three relations,

$$\sigma = \frac{F}{A_0} \qquad \varepsilon = \frac{\sigma}{E} \qquad \Delta l = \varepsilon \, l_0$$

the elongation is proportional to the initial length. The condition $\Delta l \le 0.25$ mm is then an upper bound for l_0, and the maximum length is the one for which $\Delta l = 0.25$ mm.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Area

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi \, (10.2 \text{ mm})^2}{4}$$

The square is $10.2^2 = 104.04$, then $\pi \times 104.04 = 326.85$ and dividing by 4:

$$A_0 = 81.71 \text{ mm}^2$$

2. Stress and strain at the largest elastic load

$$\sigma = \frac{F}{A_0} = \frac{8900 \text{ N}}{81.71 \text{ mm}^2} = 108.92 \text{ MPa}$$

$$\varepsilon = \frac{\sigma}{E} = \frac{108.92 \text{ N/mm}^2}{207000 \text{ N/mm}^2} = 5.2617 \times 10^{-4}$$

3. Maximum initial length

From $\Delta l = \varepsilon\, l_0$ with $\Delta l = 0.25$ mm:

$$l_0 = \frac{\Delta l}{\varepsilon} = \frac{0.25\ \text{mm}}{5.2617 \times 10^{-4}} = \mathbf{475.1\ mm}$$

The same result in one step, without the intermediate strain:

$$l_0 = \frac{\Delta l\, A_0\, E}{F} = \frac{0.25\ \text{mm} \times 81.71\ \text{mm}^2 \times 207000\ \text{N/mm}^2}{8900\ \text{N}} = 475.1\ \text{mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\Delta l\, A_0 E / F$: mm · mm² · (N/mm²) / N = mm.
- **Back substitution.** For $l_0 = 475.1$ mm, $\Delta l = 5.2617 \times 10^{-4} \times 475.1 = 0.250$ mm, the allowed elongation.
- **Elastic range.** The strain, 5.2617×10^{-4}, is of the order of 10^{-4}, typical of the elastic deformation of a metal, so the use of Hooke's law is consistent with the statement.
- **Limit.** For a shorter sample the elongation is smaller (it is proportional to l_0) and the condition is satisfied with margin; for a longer one it is exceeded.

Interpretation

The sample can be at most about 475 mm long. The limit is imposed by the stiffness E (a stiff material elongates little) and by the section (a thicker sample has less stress for the same load); it is not imposed by the strength, because the stress stays elastic.

Check against the official solution

Item	Official	Computed	Comment
l_0	475 mm	475.1 mm	agrees

Problem 7 | Problems T5_MechanicalProperties.pdf #7 | Poisson's ratio · safety factor · design

Crane Cable: Poisson's Ratio, Safety Factor and Replacement Cable

Original statement · Problems T5_MechanicalProperties.pdf, p. 3

7. In a construction site, a change in the diameter of one of the crane cables was detected. When the weight that the cable was withstanding was 48950 N, the change was 5.3 10^{-3} mm, and when the load was removed, the cable recovered its initial diameter of 15 mm. Considering that for this kind of steel the Poisson's ratio ν = 0.33 and according to the manufacturer data the initial length of the cable is 3 m.

- *a. Calculate the length of the cable when it's holding the load mentioned above.*
- *b. If it has been detected that plastic deformation of the cable begins when it is holding a weight of 10000 kg, can it be considered that the cable is working under safe conditions? Assume a safety factor of 2.*
- *c. What is the value of the elastic modulus for this material?*
- *d. In order to prevent service problems the provider replaces the cable for a different one, which has a stress-strain curve represented below and an elastic modulus of E = 75 GPa. Can this material replace the previous one? Explain why.*
- *e. Calculate the final length of this new cable in case it is working under the same conditions (48950 N).*

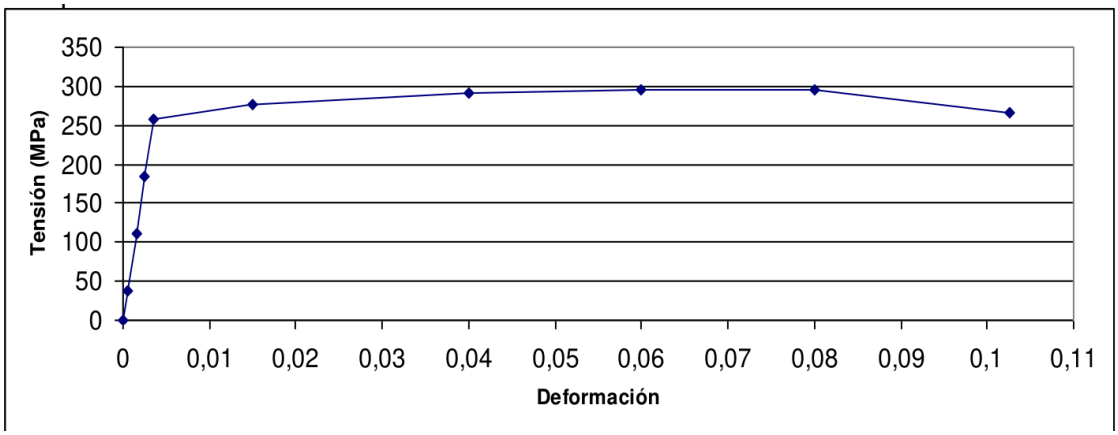


Figure 6. Problem 7: stress-strain curve of the replacement cable, copied from the sheet (Problems T5_MechanicalProperties.pdf, page 3).

Official Solution: a. $l = 3003.2$ mm; b. yes; c. $E = 261.3$ GPa; d. no; e. $l = 3045$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Elastic and isotropic cable under a uniaxial load.** When it holds the weight the cable is elastic (it recovers its diameter when the load is removed), the stress is uniform, and Hooke's law $\sigma = E\varepsilon$ and $\varepsilon_{lat} = -\nu\varepsilon_{ax}$ hold [Session 11 Slides 13 and 19].
2. **Lateral strain from the diameter.** The change of $5.3{\cdot}10^{-3}$ mm is a decrease of the diameter ($\Delta d < 0$ under tension), and the strain is relative to the initial diameter, 15 mm.
3. **Yield load.** "Plastic deformation begins when it holds a weight of 10 000 kg" means that the yield strength is the stress that this weight produces; the weight in newtons is $m\,g$ with $g = 9.81$ m/s² [Session 11 Slide 28].
4. **Safety factor.** The working stress is $\sigma_w = \sigma_y / N$, and the cable works under safe conditions if the stress that it supports does not exceed σ_w [Session 11 Slide 29].
5. **Replacement cable.** Its stress-strain curve is the one printed on the sheet (a chart of ten points), with the modulus given, $E = 75$ GPa. The yield strength of the new material is taken with the 0.2 % offset [Session 11 Slide 28], and the readings of the chart are the values of the markers.

Data

Quantity	Symbol	Value	In SI
Load in service	F	48 950 N	48 950 N
Initial diameter	d_0	15 mm	15×10^{-3} m
Change of diameter	Δd	-5.3×10^{-3} mm	-5.3×10^{-6} m
Poisson's ratio	ν	0.33	dimensionless
Initial length	l_0	3 m	3000 mm
Weight at which plastic deformation begins	m	10 000 kg	98 100 N
Safety factor	N	2	dimensionless
Modulus of the replacement cable	E_{new}	75 GPa	75 000 N/mm²

Unknowns: the length under load (a), whether the cable is safe (b), E (c), whether the new cable can replace it (d), and its length under load (e).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the cable) and the two equal transverse directions; no change of basis.

Strategy.

- (a) The measured change of the diameter gives the transverse strain; Poisson's ratio turns it into the axial strain, and the axial strain into the elongation.
- (c) Once the axial strain is known, the modulus is the stress divided by it. Item c is solved before b and d because E is used in the discussion.
- (b) The weight of 10 000 kg gives the yield strength; the safety factor gives the working stress, which is compared with the stress in service.

- (d), (e) The new cable is compared with the old one through the yield strength of the curve of the sheet: if the service stress exceeds it, the cable deforms permanently. The length (e) is read on the curve at the service stress, which is the same, since the section and the load are the same.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section and stress in service

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15\ \text{mm})^2}{4}$$

With $15^2 = 225$ and $\pi \times 225 = 706.86$:

$$A_0 = 176.71\ \text{mm}^2 \qquad \sigma = \frac{F}{A_0} = \frac{48\,950\ \text{N}}{176.71\ \text{mm}^2} = 277.00\ \text{MPa}$$

a) Length of the cable under the load

The lateral strain is the change of the diameter over the initial diameter, negative because the diameter shrinks:

$$\varepsilon_{lat} = \frac{\Delta d}{d_0} = \frac{-5.3 \times 10^{-3}\ \text{mm}}{15\ \text{mm}} = -3.5333 \times 10^{-4}$$

From $\nu = -\varepsilon_{lat}/\varepsilon_{ax}$ [Session 11 Slide 19], the axial strain is

$$\varepsilon_{ax} = -\frac{\varepsilon_{lat}}{\nu} = \frac{3.5333 \times 10^{-4}}{0.33} = 1.0707 \times 10^{-3}$$

and the elongation and the length are

$$\Delta l = \varepsilon_{ax}\, l_0 = 1.0707 \times 10^{-3} \times 3000\ \text{mm} = 3.212\ \text{mm} \qquad l = l_0 + \Delta l = 3000 + 3.212 = \mathbf{3003.2\ mm}$$

c) Elastic modulus

$$E = \frac{\sigma}{\varepsilon_{ax}} = \frac{277.00\ \mathrm{N/mm^2}}{1.0707 \times 10^{-3}} = 258708\ \mathrm{MPa} = \mathbf{258.7\ GPa}$$

b) Is the cable working under safe conditions?

The weight at which the plastic deformation begins is $W_y = 10\,000 \times 9.81 = 98\,100$ N, so the yield strength is

$$\sigma_y = \frac{W_y}{A_0} = \frac{98\,100\ \mathrm{N}}{176.71\ \mathrm{mm^2}} = 555.1\ \mathrm{MPa}$$

With the safety factor of 2 the working stress, and the corresponding load, are

$$\sigma_w = \frac{\sigma_y}{N} = \frac{555.1}{2} = 277.6\ \mathrm{MPa} \qquad F_w = \frac{W_y}{N} = \frac{98\,100}{2} = 49\,050\ \mathrm{N}$$

The stress in service, 277.00 MPa, is not larger than $\sigma_w = 277.6$ MPa (equivalently, $48950\ \mathrm{N} < 49\,050$ N), so the cable **works under safe conditions**: the safety factor really available is

$$N_{eff} = \frac{\sigma_y}{\sigma} = \frac{555.1}{277.00} = 2.004 \geq 2$$

The margin is very small, 0.20 % of the permitted load.

d) Can the new cable replace the previous one?

The curve of the sheet is a chart of ten points. Its markers are at the strains and stresses of the table of Problem 3 (for example, the point at $\varepsilon = 0.0035$ and $\sigma \approx 259$ MPa, and the maximum at ≈ 296 MPa), and those values are used as readings.

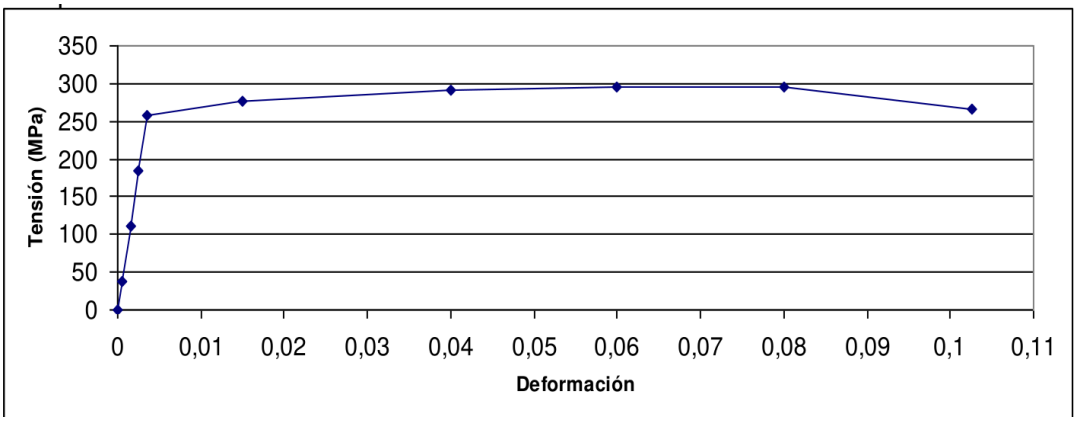


Figure 7. Problem 7: stress-strain curve of the replacement cable, copied from the sheet (Problems T5_MechanicalProperties.pdf, page 3).

The yield strength of the new material, with the offset line $E_{new}(\varepsilon - 0.002)$ and $E_{new} = 75\,000$ MPa, changes sign of $g = \sigma - E_{new}(\varepsilon - 0.002)$ between the points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.00253	184.8	39.4	145.4
4	0.00350	258.7	112.2	146.4
5	0.01496	277.2	971.8	-694.6
6	0.04002	291.9	2851.2	-2559.2

$$m = \frac{277.16 - 258.68}{0.01496 - 0.00350} = 1612.2 \text{ MPa} \qquad \varepsilon^* = \frac{258.68 - 1612.2 \times 0.00350 + 0.002 \times 75\,000}{75\,000 - 1612.2} = \frac{403.042}{73387.8} = 0.0054919$$

$$\sigma_{y,new} = E_{new}\,(\varepsilon^* - 0.002) = 75\,000 \times (0.0054919 - 0.002) = 261.9 \text{ MPa}$$

The stress that the new cable would carry is the same, 277.00 MPa, since the load and the section are the same. It is **larger** than its yield strength, $277.00 > 261.9$ MPa:

$$N_{eff,new} = \frac{\sigma_{y,new}}{\sigma} = \frac{261.9}{277.00} = 0.945 < 1 \quad \text{(the old cable has 2.004)}$$

Equivalently, the yield load of the new cable is $\sigma_{y,new} A_0 = 46.3$ kN, below the 48 950 N of service. The new cable would deform plastically and permanently under the same load, with no safety margin: the answer is **no**.

e) Length of the new cable under the same conditions

The strain is read on the curve at the service stress $\sigma = 277.00$ MPa, which is between the points 4 and 5 ($\sigma = 258.68$ MPa at $\varepsilon = 0.00350$ and $\sigma = 277.16$ MPa at $\varepsilon = 0.01496$). By linear interpolation:

$$\varepsilon = 0.00350 + \frac{277.00 - 258.68}{277.16 - 258.68}\,(0.01496 - 0.00350) = 0.01486$$

$$\Delta l = \varepsilon\, l_0 = 0.01486 \times 3000 \text{ mm} = 44.6 \text{ mm} \qquad l = 3000 + 44.6 = \mathbf{3044.6 \text{ mm}}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\Delta d/d_0$ and the strains have no units; N/mm² is MPa; $m\,g$ in kg·m/s² is N.
- **Poisson.** ε_{lat} is negative and ε_{ax} positive, and $|\varepsilon_{lat}| = 0.33\,\varepsilon_{ax}$, as the definition of ν demands.
- **Elastic range of the old cable.** The strain, 1.0707×10^{-3}, is a small fraction of the one at which it yields, $\sigma_y/E = 555.1/258708 = 2.1 \times 10^{-3}$ (it works at about half of it, as the safety factor 2 says).
- **Consistency of E.** The modulus 258.7 GPa is larger than the one of steel in the table of the slide, 207 GPa [Session 11 Slide 17], by about 25 %. The text of the sheet speaks of "this kind of steel", so the value is plausible within the data of the problem.
- **New cable.** Its length under load, 3044.6 mm, is 14 times the elongation of the old one (3.2 mm against 44.6 mm), and permanent: on unloading it recovers $\sigma/E_{new} = 277.00/75\,000$ and keeps $\varepsilon_p = 0.01117$, that is, about 33.5 mm of permanent elongation.

Interpretation

The two cables have the same section and carry the same load, but the old one works at 50 % of its yield strength and the new one above 100 %. A cable is not defined by its stiffness but by its yield strength: the new one is also much more flexible ($E = 75$ GPa against 259 GPa) and elongates by about 14 times as much. To keep the safety factor of 2 with the new material the section should be at least $A \geq NF/\sigma_{y,new} = 2 \times 48950/261.9 = 373.8$ mm², that is, a diameter of 21.8 mm instead of 15 mm.

THE LAST ITEM OF THE SHEET

The statement prints a "f." after item e, followed only by the chart of the stress-strain curve. There is no item f: the chart is the one that item d refers to ("a stress-strain curve represented below").

Check against the official solution

Item	Official	Computed	Comment
a. length	3003.2 mm	3003.2 mm	agrees
b. safe?	yes	yes ($N_{eff} = 2.004$)	agrees, with a margin of 0.20 %
c. E	261.3 GPa	258.7 GPa	differs 1 %
d. can replace?	no	no ($N_{eff,new} = 0.945$)	agrees
e. length	3045 mm	3044.6 mm	agrees

DISCREPANCY WITH THE OFFICIAL SOLUTION: MODULUS OF THE CABLE

The key gives $E = 261.3$ GPa. The data of the statement give 258.7 GPa from the unrounded strain, and 259.7 GPa if the elongation is rounded to 3.2 mm. The 261.3 GPa of the key is not reproduced by either value (it needs an axial strain of 1.060×10^{-3}); the difference is 1 % and it does not affect the other answers. The computed value is kept.

Problem 8 | Problems T5_MechanicalProperties.pdf #8 | Yield strength · Hooke's law

Elastic Limits of a Brass Sample: Stress, Load and Length without Plastic Deformation

Original statement · Problems T5_MechanicalProperties.pdf, p. 4

8. For a particular kind of brass, plastic deformation begins at 345 MPa, and its elastic modulus is 103 GPa.

>

> - a. What is the value of the yield stress?

> - b. What is the maximum stress that can be applied to a sample of this brass with a cross-section 130 mm² without plastic deformation taking place? And the maximum load?

> - c. If the initial length of the sample is 76 mm, what is the maximum length that it can reach without plastic deformation?

Official Solution: a. $\sigma_y = 345$ **MPa; b.** $\sigma_{max} = 345$ **MPa,** $F_{max} = 44850$ **N; c.** $l = 76.25$ **mm**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Yield strength.** "Plastic deformation begins at 345 MPa" means that 345 MPa is the yield strength of the brass: below it the deformation is only elastic, above it part of it is permanent [Session 11 Slide 28].
2. **Linear elastic below the yield strength**, $\sigma = E\varepsilon$ [Session 11 Slide 13], with the given modulus.
3. **Uniaxial, uniform stress**: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9]. The maximum values without plastic deformation are those that correspond to the stress σ_y.

Data

Quantity	Symbol	Value	In SI
Stress at which plastic deformation begins	σ_y	345 MPa	345×10^6 Pa
Young's modulus	E	103 GPa	103 000 N/mm²
Cross-section (item b)	A_0	130 mm²	130×10^{-6} m²
Initial length (item c)	l_0	76 mm	76×10^{-3} m

Unknowns: σ_y (a), the maximum stress and load (b), the maximum length (c).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), no change of basis.

Strategy. The three items are the same limit seen from three sides. The largest stress that does not cause plastic deformation is the yield strength (a, b). The largest load is that stress times the area, $F = \sigma_y A_0$ (b). The largest length is the initial length plus the elongation at the yield strength, and that elongation comes from Hooke's law, $\varepsilon_y = \sigma_y / E$ and $\Delta l = \varepsilon_y \, l_0$ (c).

PHASE 3 · STEP-BY-STEP DERIVATION

a) Yield stress

By the hypothesis, plastic deformation begins when $\sigma = 345$ MPa, so

$$\sigma_y = \mathbf{345\ MPa}$$

b) Maximum stress and maximum load

Without plastic deformation the stress cannot be larger than the yield strength, so

$$\sigma_{max} = \sigma_y = \mathbf{345\ MPa}$$

The load that produces that stress in a section of 130 mm² follows from $\sigma = F / A_0$:

$$F_{max} = \sigma_{max}\, A_0 = 345\ \frac{\mathrm{N}}{\mathrm{mm}^2} \times 130\ \mathrm{mm}^2 = 44\,850\ \mathrm{N} = \mathbf{44.85\ kN}$$

c) Maximum length without plastic deformation

The strain at the yield strength comes from Hooke's law:

$$\varepsilon_y = \frac{\sigma_y}{E} = \frac{345\ \mathrm{N/mm}^2}{103000\ \mathrm{N/mm}^2} = 3.3495 \times 10^{-3}$$

The elongation of a sample of 76 mm is

$$\Delta l = \varepsilon_y \, l_0 = 3.3495 \times 10^{-3} \times 76 \text{ mm} = 0.2546 \text{ mm}$$

and the maximum length that it can reach without plastic deformation is

$$l = l_0 + \Delta l = 76 + 0.2546 = \mathbf{76.255 \text{ mm}}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** (N/mm²)·mm² = N for the load; (N/mm²)/(N/mm²) has no units for the strain; strain × mm = mm for the elongation.
- **Order of magnitude.** $\varepsilon_y = 0.335$ %, a few tenths of a percent, the order of magnitude of the elastic strain of a metal.
- **Limit.** If the load were zero, $l = l_0 = 76$ mm. At the yield load, the sample is only 0.25 mm longer, that is, 0.33 % of its length.

Interpretation

Up to 44.85 kN the sample returns to 76 mm when unloaded. Beyond that load part of the elongation is permanent. The brass elongates very little (a quarter of a millimetre) before it yields: the yield strength is a stress limit, but it translates into a small strain because E is large.

Check against the official solution

Item	Official	Computed	Comment
a. σ_y	345 MPa	345 MPa	agrees
b. σ_{max}, F_{max}	345 MPa, 44 850 N	345 MPa, 44\,850 N	agrees
c. l	76.25 mm	76.25 mm	agrees

`Problem 9` `Problems T5_MechanicalProperties.pdf #9` `Hooke's law · Poisson's ratio`

Elongation and Change of Diameter of a Steel Sample under 48 900 N

`Original statement · Problems T5_MechanicalProperties.pdf, p. 4`

9. A load of 48900 N produces elastic deformation in a cylindrical sample of 15.2 mm in diameter and a length of 254 mm. Considering that for this kind of steel E = 207 GPa and ν = 0.27, determine the elongation in the direction of the applied force and the change in diameter of the sample.

Official Solution: $\Delta l = 0.33$ mm, $\Delta\phi = -0.0053$ mm

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Elastic deformation.** The statement says that the load produces only elastic deformation, so Hooke's law holds, $\sigma = E\varepsilon$ [Session 11 Slide 13].
2. **Uniaxial, uniform stress** along the axis of the cylinder: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
3. **Isotropic material.** With a uniaxial stress, the transverse strain is $\varepsilon_{lat} = -\nu\,\varepsilon_{ax}$, equal in the two transverse directions [Session 11 Slide 19].
4. The cross-section is circular and uniform, so that the diameter changes in the same proportion in every direction of the section.

Data

Quantity	Symbol	Value	In SI
Load	F	48 900 N	48 900 N
Diameter	d_0	15.2 mm	15.2×10^{-3} m
Length	l_0	254 mm	254×10^{-3} m
Young's modulus	E	207 GPa	207 000 N/mm²
Poisson's ratio	ν	0.27	

Unknowns: the elongation Δl in the direction of the force and the change of the diameter Δd.

`PHASE 2 · FRAMES & STRATEGY`

One direction for the load (the axis) and the two equal transverse directions of the cylinder; no change of basis.

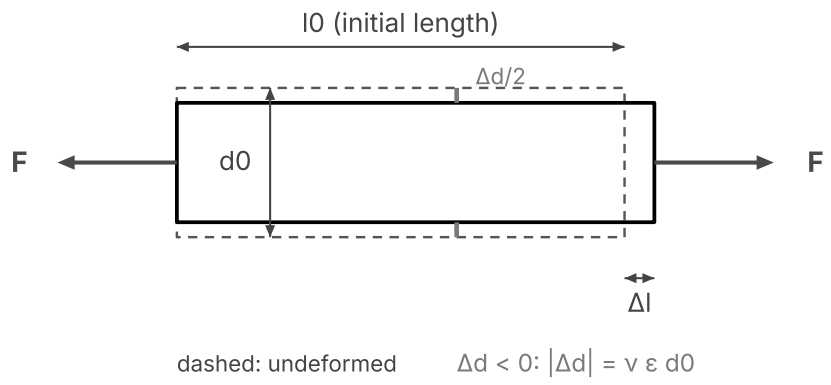


Figure 8. Cylindrical bar under an axial load F: it elongates by Δl and its diameter shrinks by Δd (schematic, exaggerated).

Strategy. The elongation follows from the stress and the modulus, and the contraction of the diameter follows from the same strain through Poisson's ratio [Session 11 Slide 19]:

$$\sigma = \frac{F}{A_0} \qquad \varepsilon_{ax} = \frac{\sigma}{E} \qquad \Delta l = \varepsilon_{ax}\, l_0 \qquad \varepsilon_{lat} = -\nu\, \varepsilon_{ax} \qquad \Delta d = \varepsilon_{lat}\, d_0$$

The minus sign says that the diameter decreases when the sample is stretched.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15.2\ \text{mm})^2}{4}$$

The square is $15.2^2 = 231.04$, then $\pi \times 231.04 = 725.83$ and dividing by 4:

$$A_0 = 181.46\ \text{mm}^2$$

2. Stress and axial strain

$$\sigma = \frac{F}{A_0} = \frac{48\,900\ \text{N}}{181.46\ \text{mm}^2} = 269.48\ \text{MPa}$$

$$\varepsilon_{ax} = \frac{\sigma}{E} = \frac{269.48\ \text{N/mm}^2}{207000\ \text{N/mm}^2} = 1.3019 \times 10^{-3}$$

3. Elongation in the direction of the force

$$\Delta l = \varepsilon_{ax}\, l_0 = 1.3019 \times 10^{-3} \times 254\ \text{mm} = \mathbf{0.331\ mm}$$

4. Change of the diameter

The transverse strain is

$$\varepsilon_{lat} = -\nu\, \varepsilon_{ax} = -0.27 \times 1.3019 \times 10^{-3} = -3.5150 \times 10^{-4}$$

and the change of the diameter is

$$\Delta d = \varepsilon_{lat}\, d_0 = -3.5150 \times 10^{-4} \times 15.2\ \text{mm} = \mathbf{-0.00534\ mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** (N/mm²)/(N/mm²) has no units for the strains, and strain × mm gives mm for Δl and Δd.
- **Elastic range.** $\varepsilon_{ax} = 0.1302$ %, a small strain of the order of those of the elastic deformation of a metal, consistent with the statement.
- **Volume.** The relative change of volume is $\Delta V/V \approx \varepsilon_{ax} + 2\varepsilon_{lat} = \varepsilon_{ax}(1 - 2\nu) = 5.989 \times 10^{-4}$, positive: a stretched elastic body increases its volume since $\nu < 0.5$ (a perfectly incompressible material, $\nu = 0.5$, would give zero) [Session 11 Slide 19].
- **Sign.** $\Delta l > 0$ and $\Delta d < 0$: the sample gets longer and thinner.

Interpretation

The sample elongates by 0.331 mm, about a third of a millimetre in 254 mm, and the diameter shrinks by only 5.34 micrometres. The contraction is a fraction $\nu = 0.27$ of the relative elongation, which is why it is so small and why it is difficult to measure directly.

Check against the official solution

Item	Official	Computed	Comment
Δl	0.33 mm	0.331 mm	agrees
Δd	−0.0053 mm	-0.0053 mm	agrees

`Problem 10` `Problems T5_MechanicalProperties.pdf #10` `Yield · unloading · resilience`

Copper Wire: Yield Strength, Permanent Elongation after Unloading and Resilience

`Original statement · Problems T5_MechanicalProperties.pdf, p. 4`

10. Plastic deformation in a copper wire of 200 cm in length and a diameter of 1 mm begins when it is loaded with 15.3 kg. If the elastic modulus for copper is 110 GPa.

- *a. Calculate its yield strength.*
- *b. Considering that when it is loaded with 29.6 kg the strain is 0.011, calculate the final length of the wire once it has been unloaded.*
- *c. Calculate the resilience modulus.*

Official Solution: a. $\sigma_y = 190.9$ **MPa; b.** $l = 201.85$ **cm; c.** $U_r = 0.165$ **MPa**

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Uniaxial, uniform stress** in the wire: $\sigma = F/A_0$ [Session 11 Slide 9]. The weight of a mass m is $F = m\,g$ with $g = 9.81$ m/s².
2. **Yield strength.** "Plastic deformation begins when it is loaded with 15.3 kg" gives the yield strength, $\sigma_y = F_y/A_0$ [Session 11 Slide 28].
3. **Total strain under load.** The strain 0.011 of item b is the total strain (elastic plus plastic) measured while the load of 29.6 kg is applied, relative to the initial length: $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
4. **Unloading is elastic.** When the load is removed the elastic strain is recovered along a line of slope E, parallel to the elastic line, starting from the stress that the wire carried; what remains is the plastic strain [Session 11 Slides 21 and 32].
5. **Resilience** of the elastic range: $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35].

Data

Quantity	Symbol	Value	In SI
Length	l_0	200 cm	2.000 m = 2000 mm
Diameter	d_0	1 mm	1×10^{-3} m
Young's modulus	E	110 GPa	110 000 N/mm²
Mass at which plastic deformation begins	m_1	15.3 kg	$F_1 = 15.3 \times 9.81 = 150.09$ N
Mass of item b	m_2	29.6 kg	$F_2 = 29.6 \times 9.81 = 290.38$ N
Total strain under m_2	ε	0.011	0.011 (dimensionless)

Unknowns: σ_y (a), the final length after unloading (b), U_r (c).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the wire); no change of basis.

Strategy. (a) is a stress: weight over area. (c) follows from (a) and E. Item (b) is the delicate one, and it is solved by following the strain along the load path:

1. The wire is loaded up to $\sigma_2 = F_2/A_0$, above the yield strength, and the total strain there is 0.011.
2. It is unloaded along a straight line of slope E: the strain recovered is σ_2/E (the stress that it carried, not the yield strength).
3. The strain that remains is permanent, and it gives the final length.

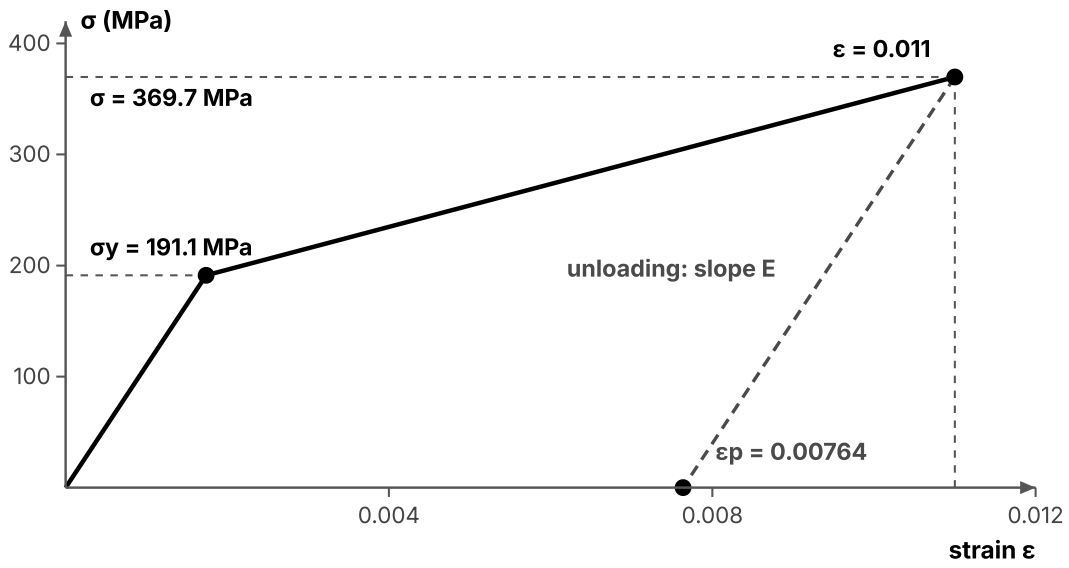


Figure 9. Problem 10: loading beyond the yield point and unloading along a line of slope E; the strain that remains is permanent (the shape between the yield point and the loaded point is not given by the statement).

PHASE 3 · STEP-BY-STEP DERIVATION

1. Area of the wire

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(1\ \text{mm})^2}{4} = 0.7854\ \text{mm}^2$$

a) Yield strength

The force at which the plastic deformation begins is

$$F_1 = m_1\, g = 15.3\ \text{kg} \times 9.81\ \text{m/s}^2 = 150.09\ \text{N}$$

$$\sigma_y = \frac{F_1}{A_0} = \frac{150.09\ \text{N}}{0.7854\ \text{mm}^2} = \mathbf{191.1\ MPa}$$

b) Length after unloading

The stress under the load of 29.6 kg is

$$F_2 = 29.6 \times 9.81 = 290.38\ \text{N} \qquad \sigma_2 = \frac{290.38}{0.7854} = 369.7\ \text{MPa}$$

which is above the yield strength (93 % higher), so the wire is in the plastic zone and the total strain, $\varepsilon = 0.011$, is the sum of an elastic part and a plastic part, $\varepsilon = \varepsilon_e + \varepsilon_p$. On unloading, the elastic part is recovered along the line of slope E that starts at σ_2:

$$\varepsilon_e = \frac{\sigma_2}{E} = \frac{369.7\ \text{N/mm}^2}{110000\ \text{N/mm}^2} = 3.3611 \times 10^{-3}$$

The strain that remains is

$$\varepsilon_p = \varepsilon - \varepsilon_e = 0.011 - 3.3611 \times 10^{-3} = 0.007639$$

and the final length, once unloaded, is

$$l_f = l_0\,(1 + \varepsilon_p) = 200\ \text{cm} \times (1 + 0.007639) = \mathbf{201.53\ cm}$$

(that is, the wire is left 1.53 cm longer than at the beginning).

c) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(191.1\ \text{MPa})^2}{2 \times 110000\ \text{MPa}} = \mathbf{0.166\ MPa}$$

($1\ \text{MPa} = 1\ \text{MJ/m}^3$.)

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** kg·m/s² = N; N/mm² = MPa; the strains have no units; l_f in the unit of l_0 (cm).
- **Order.** $\varepsilon_e = 3.3611 \times 10^{-3} < \varepsilon = 0.011$, and $\varepsilon_p > 0$: the wire does not return to 200 cm. If the load were removed from the yield point, the permanent strain would be zero; here the wire has gone beyond it.
- **Elastic strain at yield.** The strain of the yield point, $\sigma_y/E = 0.001737$, is smaller than the elastic strain recovered from σ_2 (3.3611×10^{-3}): the unloading starts from a higher stress, and the recovery is larger.
- **Limits.** If $\sigma_2 = \sigma_y$ the wire would be exactly at the yield point and ε_p would be (almost) zero; the larger the stress reached, the larger the elastic recovery and the permanent strain.

Interpretation

The copper wire yields at about 191.1 MPa, and after being loaded to 369.7 MPa it keeps a permanent elongation of 1.53 cm in 200 cm. A load that goes beyond the yield strength leaves a trace: the length after unloading depends on how far the stress went.

VALUE OF g

The key is consistent with $g = 9.8$ m/s², which gives $\sigma_y = 190.9$ MPa instead of 191.1 MPa (0.1 % difference) and $U_r = 0.1657$ MPa. The result here uses $g = 9.81$ m/s² throughout the sheet.

Check against the official solution

Item	Official	Computed	Comment
a. σ_y	190.9 MPa	191.1 MPa	agrees (the key uses $g = 9.8$ m/s²)
b. final length	201.85 cm	201.53 cm	**does not agree**, see below
c. U_r	0.165 MPa	0.166 MPa	agrees

DISCREPANCY WITH THE OFFICIAL SOLUTION: LENGTH AFTER UNLOADING

The key, 201.85 cm, is obtained by recovering only the elastic strain of the yield point, $\sigma_y/E = 1.7373 \times 10^{-3}$: $l = 200\,(1 + 0.011 - 1.7373 \times 10^{-3}) = 201.85$ cm. That is the unloading of a wire that had been stressed to σ_y. But the wire of item b carried $\sigma_2 = 369.7$ MPa, and on unloading it recovers $\sigma_2/E = 3.3611 \times 10^{-3}$. The correct permanent strain is therefore 0.007639 and the length 201.53 cm. The key therefore recovers less elastic strain than the wire really does. The computed value is kept.

Problem 11 | Problems T5_MechanicalProperties.pdf #11 | Hooke's law · Poisson's ratio

Elastic Deformation of a 15 mm Steel Cylinder under 50 kN

Original statement · Problems T5_MechanicalProperties.pdf, p. 4

11. A cylindrical steel sample with a diameter of 15 mm and a length of 250 mm experiences elastic deformation when a load of 50 kN is applied. Considering that for this steel E = 207 GPa and ν = 0.27, determine the elongation in the direction of the force applied and the change in diameter of the sample.

Official Solution: $\Delta l = 0.342$ mm, $\Delta\phi = -0.0055$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Elastic deformation.** The statement says that the load produces only elastic deformation, so Hooke's law holds, $\sigma = E\varepsilon$ [Session 11 Slide 13].
2. **Uniaxial, uniform stress** along the axis of the cylinder: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
3. **Isotropic material.** With a uniaxial stress, the transverse strain is $\varepsilon_{lat} = -\nu\,\varepsilon_{ax}$, equal in the two transverse directions [Session 11 Slide 19].
4. The cross-section is circular and uniform, so that the diameter changes in the same proportion in every direction of the section.

Data

Quantity	Symbol	Value	In SI
Load	F	50 000 N	50 000 N
Diameter	d_0	15.0 mm	15.0×10^{-3} m
Length	l_0	250 mm	250×10^{-3} m
Young's modulus	E	207 GPa	207 000 N/mm²
Poisson's ratio	ν	0.27	

Unknowns: the elongation Δl in the direction of the force and the change of the diameter Δd.

PHASE 2 · FRAMES & STRATEGY

One direction for the load (the axis) and the two equal transverse directions of the cylinder; no change of basis.

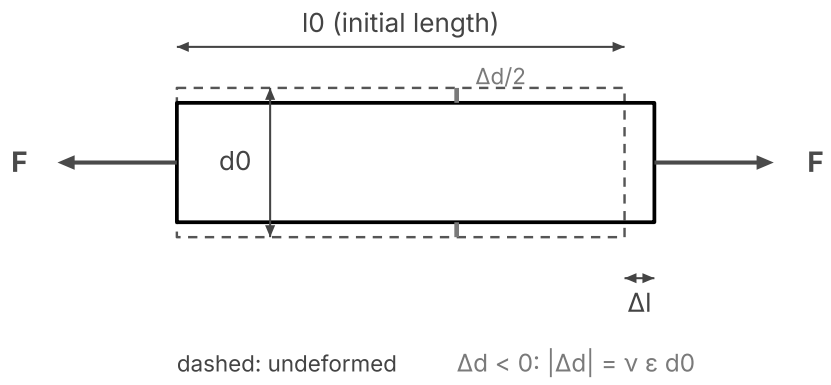


Figure 10. Cylindrical bar under an axial load F: it elongates by Δl and its diameter shrinks by Δd (schematic, exaggerated).

Strategy. The elongation follows from the stress and the modulus, and the contraction of the diameter follows from the same strain through Poisson's ratio [Session 11 Slide 19]:

$$\sigma = \frac{F}{A_0} \qquad \varepsilon_{ax} = \frac{\sigma}{E} \qquad \Delta l = \varepsilon_{ax}\, l_0 \qquad \varepsilon_{lat} = -\nu\, \varepsilon_{ax} \qquad \Delta d = \varepsilon_{lat}\, d_0$$

The minus sign says that the diameter decreases when the sample is stretched.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15.0\ \text{mm})^2}{4}$$

The square is $15.0^2 = 225.00$, then $\pi \times 225.00 = 706.86$ and dividing by 4:

$$A_0 = 176.71\ \text{mm}^2$$

2. Stress and axial strain

$$\sigma = \frac{F}{A_0} = \frac{50\,000\ \mathrm{N}}{176.71\ \mathrm{mm}^2} = 282.94\ \mathrm{MPa}$$

$$\varepsilon_{ax} = \frac{\sigma}{E} = \frac{282.94\ \mathrm{N/mm}^2}{207000\ \mathrm{N/mm}^2} = 1.3669 \times 10^{-3}$$

3. Elongation in the direction of the force

$$\Delta l = \varepsilon_{ax}\, l_0 = 1.3669 \times 10^{-3} \times 250\ \mathrm{mm} = \mathbf{0.342\ mm}$$

4. Change of the diameter

The transverse strain is

$$\varepsilon_{lat} = -\nu\, \varepsilon_{ax} = -0.27 \times 1.3669 \times 10^{-3} = -3.6905 \times 10^{-4}$$

and the change of the diameter is

$$\Delta d = \varepsilon_{lat}\, d_0 = -3.6905 \times 10^{-4} \times 15.0\ \mathrm{mm} = \mathbf{-0.00554\ mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** (N/mm²)/(N/mm²) has no units for the strains, and strain × mm gives mm for Δl and Δd.
- **Elastic range.** $\varepsilon_{ax} = 0.1367$ %, a small strain of the order of those of the elastic deformation of a metal, consistent with the statement.
- **Volume.** The relative change of volume is $\Delta V/V \approx \varepsilon_{ax} + 2\varepsilon_{lat} = \varepsilon_{ax}(1 - 2\nu) = 6.288 \times 10^{-4}$, positive: a stretched elastic body increases its volume since $\nu < 0.5$ (a perfectly incompressible material, $\nu = 0.5$, would give zero) [Session 11 Slide 19].
- **Sign.** $\Delta l > 0$ and $\Delta d < 0$: the sample gets longer and thinner.

Interpretation

The sample elongates by 0.342 mm, about a third of a millimetre in 250 mm, and the diameter shrinks by only 5.54 micrometres. The contraction is a fraction $\nu = 0.27$ of the relative elongation, which is why it is so small and why it is difficult to measure directly.

Check against the official solution

Item	Official	Computed	Comment
Δl	0.342 mm	0.342 mm	agrees
Δd	−0.0055 mm	-0.0055 mm	agrees

Problem 12 | Problems T5_MechanicalProperties.pdf #12 | True stress · true strain

Steel Cable: True Fracture Stress and True Strain from the Reduction in Area

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

12. A steel cable has fracture strength of 300 MPa. Its ductility is 77% measured as a reduction in area. Calculate the real fracture strength. Considering that strain is uniform until fracture, what is the real strain?

Official Solution: $\varepsilon_{real} = 1.47$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Fracture strength = engineering stress at fracture.** The "fracture strength of 300 MPa" is the load at fracture divided by the *initial* area, $\sigma_f = F_f / A_0$, as in the engineering curve of the tensile test [Session 11 Slide 9]. The true fracture stress is that load divided by the area at fracture, A_f [Session 11 Slide 37].
2. **Ductility as reduction in area.** $RA\,(\%) = \dfrac{A_0 - A_f}{A_0} \times 100$ is measured after fracture [Session 11 Slide 33], so $RA = 77\,\%$ means $A_f = 0.23\,A_0$.
3. **Uniform deformation.** The statement asks to take the strain as uniform until fracture, so the volume of the wire is constant, $l_0 A_0 = l_f A_f$ (the necking is not modelled) [Session 11 Slide 38].
4. **Real strain by integration.** $\varepsilon_r = \int_{l_0}^{l} dl/l = \ln(l/l_0)$ [Session 11 Slide 37].

Data

Quantity	Symbol	Value	In SI
Engineering fracture strength	σ_f	300 MPa	300×10^6 Pa
Reduction in area	RA	77 %	0.77 (dimensionless)

Unknowns: the true fracture stress $\sigma_{f,r}$ and the true strain at fracture ε_r.

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the wire); no change of basis.

Strategy.

1. The reduction in area gives the ratio $A_f/A_0 = 1 - RA$.
2. The load at fracture is the same in both definitions of stress, $F_f = \sigma_f A_0 = \sigma_{f,r} A_f$, so $\sigma_{f,r} = \sigma_f\, A_0/A_f$ (the true stress is larger than the engineering one because the area has decreased) [Session 11 Slide 38].
3. With constant volume, the length ratio is the inverse of the area ratio, and the true strain follows from $\ln(l/l_0) = \ln(A_0/A_f)$.
4. The result is verified with the other route, $\sigma_r = \sigma(1+\varepsilon)$ and $\varepsilon_r = \ln(1+\varepsilon)$, through the engineering strain.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Area ratio at fracture

$$\frac{A_f}{A_0} = 1 - \frac{RA}{100} = 1 - 0.77 = 0.23$$

2. True fracture stress

The load at fracture is $F_f = \sigma_f A_0$. Dividing by the area at fracture instead of the initial one:

$$\sigma_{f,r} = \frac{F_f}{A_f} = \frac{\sigma_f A_0}{A_f} = \frac{\sigma_f}{A_f/A_0} = \frac{300\ \text{MPa}}{0.23} = \mathbf{1304.3\ MPa}$$

($\sigma_{f,r}/\sigma_f = 4.348$: the true stress is more than four times the engineering one.)

3. True strain at fracture

With the volume constant, $l_0 A_0 = l_f A_f$, so

$$\frac{l_f}{l_0} = \frac{A_0}{A_f} = \frac{1}{0.23} = 4.348$$

and the integral of the definition of the real strain, with the substitution $u = l$, $du = dl$, and Barrow's rule between l_0 and l_f, is

$$\varepsilon_r = \int_{l_0}^{l_f} \frac{dl}{l} = \left[\ln l\right]_{l_0}^{l_f} = \ln l_f - \ln l_0 = \ln \frac{l_f}{l_0} = \ln \frac{A_0}{A_f} = \ln \frac{1}{0.23} = \mathbf{1.4697}$$

Equivalently, in terms of the diameters of the wire, $\varepsilon_r = 2\ln(D_0/D_f)$ [Session 11 Slide 38], with $D_f/D_0 = \sqrt{A_f/A_0} = \sqrt{0.23} = 0.4796$: $2\ln(1/0.4796) = 1.4697$.

4. Check through the engineering strain

The engineering strain at fracture is $\varepsilon = l_f/l_0 - 1 = 4.348 - 1 = 3.348$ (that is, 334.8 %). Then

$$\sigma_{f,r} = \sigma_f\,(1+\varepsilon) = 300 \times 4.348 = 1304.3\ \text{MPa} \qquad \varepsilon_r = \ln(1+\varepsilon) = \ln 4.348 = 1.4697$$

in agreement with the two results above.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** Stress in MPa; A_f/A_0 and the strains are dimensionless; the argument of the logarithm is a ratio of areas.
- **Limits.** If $RA \to 0$ (no necking) then $A_f \to A_0$, $\sigma_{f,r} \to \sigma_f$ and $\varepsilon_r \to 0$. As $RA \to 100\,\%$, both $\sigma_{f,r}$ and ε_r grow without bound.
- **Ordering.** $\sigma_{f,r} > \sigma_f$ and $\varepsilon_r < \varepsilon$ ($1.4697 < 3.348$), as stated in [Session 11 Slide 37].

Interpretation

The steel cable breaks when the material that is left carries 1304 MPa, not 300 MPa: the engineering value hides the reduction of the section. The true strain, 1.47, is the natural measure of such a large deformation, because the integral adds up the increments dl/l over the changing length.

LIMITATION OF THE UNIFORM-STRAIN HYPOTHESIS

A real cable necks before fracture, and most of the strain is concentrated in the neck. The mean strain along the whole length, obtained here, is therefore a lower bound of the local strain at the fracture section. The statement asks explicitly for the uniform strain, so that is the value reported.

Check against the official solution

Item	Official	Computed	Comment
true fracture stress	not given	1304.3 MPa	asked by the statement, absent from the key
true strain	1.47	1.4697	agrees

DISCREPANCY WITH THE OFFICIAL SOLUTION: THE KEY OMITS THE TRUE STRESS

The key only prints "$\varepsilon_{real} = 1.47$", although the statement asks for the real fracture strength first. The value computed here, $\sigma_{f,r} = 1304.3$ MPa, is not contradicted by the key: it is the first half of the answer that the key leaves out.

Problem 13 | Problems T5_MechanicalProperties.pdf #13 | Vickers · Brinell hardness

Vickers and Brinell Hardness Numbers from Indentation Measurements

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

13. Calculate the Vickers and Brinell hardness numbers for a metallic material on which several hardness tests were performed, obtaining the following results:

- *a. When the diamond Vickers hardness test with a load of 30 kg was performed, the average length of the diagonal was: first impression 0.527 mm, second impression 0.481 mm, third impression 0.497 mm.*
- *b. When using the Brinell test with a ball of 10 mm and a load of 3000 kg, the diameter of the indentation was 4.01 mm.*

Official Solution: 220 HV, 227 HB

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Vickers.** The indenter is a diamond right pyramid with 136° between opposite faces; the hardness is the load divided by the lateral area of the impression, $HV = F/A_{imp}$, which depends only on the mean diagonal d [Session 12 Slide 26]. The slide defines d as the arithmetic mean of the two diagonals d_1 and d_2 of one impression; here the sheet gives one diagonal length for each of the three impressions, which are averaged first.
2. **Brinell.** The hardness is the load divided by the area S of the spherical cap left by the ball, $HB = P/S$ with P in kg [Session 12 Slide 19]. The result is a number with no units attached by convention.
3. The tests are performed under their standard conditions (flat surface, indentation far from edges and from other impressions) [Session 12 Slide 28].

Data

Quantity	Symbol	Value
Vickers load	F	30 kgf (HV30)
Mean diagonal, impression 1	d_1	0.527 mm
Mean diagonal, impression 2	d_2	0.481 mm
Mean diagonal, impression 3	d_3	0.497 mm
Brinell load	P	3000 kg
Ball diameter	D	10 mm
Diameter of the indentation	D_i	4.01 mm

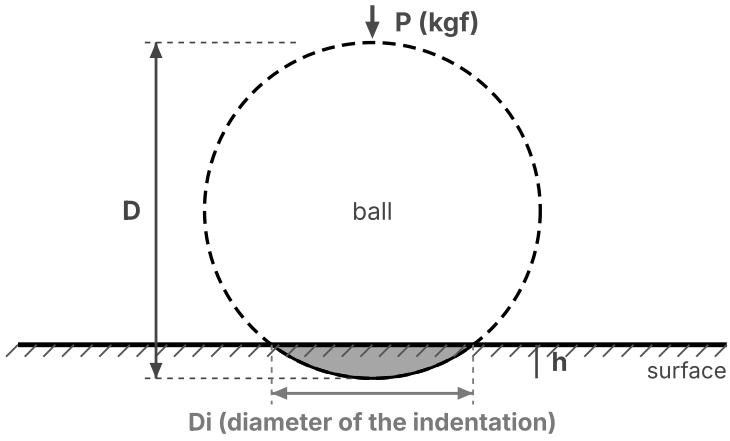
Unknowns: HV (a) and HB (b). Both loads are kilogram-force, the unit used by the slides; the hardness numbers are quoted without units (kgf/mm² in the definition, $1\ \text{kgf} = 9.81\ \text{N}$).

PHASE 2 · FRAMES & STRATEGY

No frames are involved. Each hardness comes from the load and one measured length, through the formulas of the slides.

Strategy.

- (a) Average the three diagonals, as the slide does with d_1 and d_2, and use $HV = 2F\sin(136^\circ/2)/d^2 = 1.854\,F/d^2$. The expression is derived from the pyramid: the faces make $136°$ with each other, so the lateral area is $A = d^2/(2\sin 68^\circ)$ and $HV = F/A$.
- (b) Use $HB = 2P/\left[\pi D\left(D - \sqrt{D^2 - D_i^2}\right)\right]$, which is P over the area of the cap, $S = \pi D h$ with $h = (D - \sqrt{D^2 - D_i^2})/2$ the depth of the indentation.



Cap area S = π D h with h = (D − √(D² − Di²)) / 2 ⇒ HB = P / S

Figure 11. Brinell test: the cap of height h left by the ball of diameter D (the proportions are exaggerated, not to scale).

PHASE 3 · STEP-BY-STEP DERIVATION

a) Vickers hardness

Mean diagonal.

$$\bar{d} = \frac{d_1 + d_2 + d_3}{3} = \frac{0.527 + 0.481 + 0.497}{3} = \frac{1.505}{3} = 0.50167\ \text{mm}$$

Constant of the pyramid. $\sin 68^\circ = 0.927184$, so

$$2\sin\frac{136^\circ}{2} = 2\sin 68^\circ = 1.8544$$

Hardness.

$$HV = \frac{2F\sin(136^\circ/2)}{\bar{d}^2} = \frac{1.8544 \times 30}{(0.5017)^2} = \frac{55.631}{0.251669} = \mathbf{221.0\ HV30}$$

so the hardness is about **221 HV**.

b) Brinell hardness

The indentation is characterised by its diameter $D_i = 4.01$ mm, with the ball of $D = 10$ mm:

$$D_i^2 = 4.01^2 = 16.0801 \qquad D^2 - D_i^2 = 100 - 16.0801 = 83.9199\ \text{mm}^2 \qquad \sqrt{D^2 - D_i^2} = 9.16078\ \text{mm}$$

$$D - \sqrt{D^2 - D_i^2} = 10 - 9.16078 = 0.83922\ \text{mm} \qquad \pi D\left(D - \sqrt{D^2 - D_i^2}\right) = \pi \times 10 \times 0.83922 = 26.3649\ \text{mm}^2$$

The depth of the indentation is $h = 0.83922/2 = 0.4196$ mm and the area of the cap $S = 26.3649/2 = 13.182$ mm². Therefore

$$HB = \frac{2P}{\pi D\left(D - \sqrt{D^2 - D_i^2}\right)} = \frac{2 \times 3000}{26.3649} = \mathbf{227.6\ HB}$$

that is, about **228 HB** (equivalently $P/S = 3000/13.182 = 227.6$).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** F in kgf and d in mm give kgf/mm², the unit of hardness; $D^2 - D_i^2$ in mm², and the Brinell denominator is an area (mm²), so P/S is again kgf/mm².

- **Load factor.** $Q = P/D^2 = 3000/10^2 = 30$, one of the standard values of the table of [Session 12 Slide 20] for the ball of 10 mm.
- **Order of magnitude.** $HV \approx 221$ and $HB \approx 228$ are of the same order, since both scales are defined as a load in kgf over an area in mm²; but the slides remind us that there is no universal conversion between methods [Session 12 Slide 15].
- **Limit.** If $D_i \to 0$ then $D - \sqrt{D^2 - D_i^2} \to D_i^2/(2D)$ and $HB \to 4P/(\pi D_i^2)$, the load over the area of the circle of the indentation, as it should for a very shallow cap.

Interpretation

The metal is a medium-hard alloy, of the order of 220 kgf/mm². The three Vickers impressions differ by 10 % in their diagonals (0.481 to 0.527 mm), which changes the hardness by about 20 % between impressions (240.5 to 200.3 HV): this is the scatter of the measurement, and the reason why several tests are averaged.

> **AVERAGE OF THE DIAGONALS, NOT OF THE HARDNESS NUMBERS**
>
> The slide defines d as the mean of the diagonals of the impression, so the diagonals are averaged first and one hardness is computed. Averaging the three hardness numbers instead (225.2, 240.5, 200.3) would give 222.0 HV, which is not the value defined by the slide.

Check against the official solution

Item	Official	Computed	Comment
Vickers	220 HV	221.0 HV	agrees to the rounding of the key
Brinell	227 HB	227.6 HB	agrees (the key truncates)

> **SMALL DIFFERENCE WITH THE OFFICIAL SOLUTION: VICKERS**
>
> The key gives 220 HV and the computation, with 1.854 from $2\sin 68^\circ = 1.8544$, gives 221.0 HV. The difference is below 0.5 %. The computed value is kept; the key looks rounded down to the tens.

Problem 14 | Problems T5_MechanicalProperties.pdf #14 | Brinell hardness

Brinell Hardness of a Steel from the Indentation

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

14. A Brinell indenter of 10 mm in diameter produced an indentation of 2.5 mm in diameter in a steel when a load of 1000 kg was applied. Calculate the Brinell hardness for that material.

Official Solution: 200 HB

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. The indentation is a spherical cap left by a hard ball of diameter D pressed with the load P, and the hardness is $HB = P/S$ [Session 12 Slide 19].
2. P is in kilogram-force (kg in the slide) and the lengths in mm, so HB is in kgf/mm² and is quoted as a number.
3. The test is standard: the load factor $Q = P/D^2$ of the test must be one of those of [Session 12 Slide 20].

Data

Quantity	Symbol	Value
Ball diameter	D	10 mm
Diameter of the indentation	D_i	2.5 mm
Load	P	1000 kg

Unknown: the Brinell hardness HB.

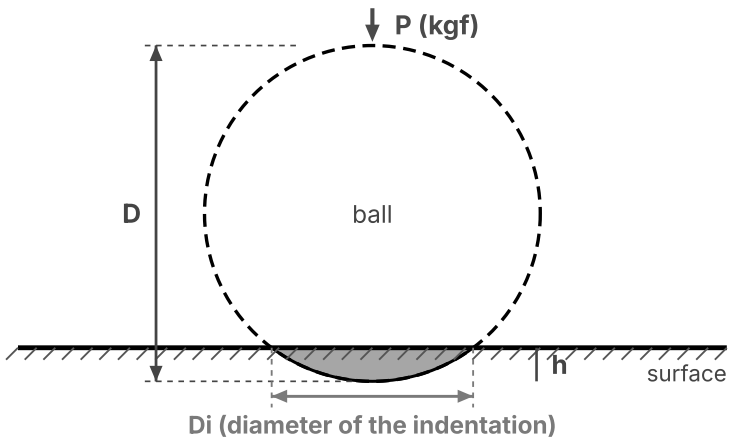
PHASE 2 · FRAMES & STRATEGY

No frames are involved: only one formula, applied directly.

Why this formula. The slide writes $HB = P/S$ with S the area of the spherical cap. For a sphere of diameter D, the cap of height h has area $S = \pi D h$ (Archimedes), and its base circle has radius $D_i/2$, so h follows from $(D/2 - h)^2 + (D_i/2)^2 = (D/2)^2$:

$$h = \frac{D - \sqrt{D^2 - D_i^2}}{2} \quad \Rightarrow \quad S = \frac{\pi D}{2}\left(D - \sqrt{D^2 - D_i^2}\right) \quad \Rightarrow \quad HB = \frac{2P}{\pi D\left(D - \sqrt{D^2 - D_i^2}\right)}$$

which is the expression of [Session 12 Slide 19].



Cap area S = π D h with h = (D − √(D² − Di²)) / 2 ⇒ HB = P / S

Figure 12. Brinell test: the cap of height h left by the ball of diameter D (the proportions are exaggerated, not to scale).

PHASE 3 · STEP-BY-STEP DERIVATION

Square of the indentation diameter and of the ball diameter:

$$D_i^2 = 2.5^2 = 6.25 \text{ mm}^2 \qquad D^2 = 10^2 = 100 \text{ mm}^2$$

Difference and square root:

$$D^2 - D_i^2 = 100 - 6.25 = 93.75 \text{ mm}^2 \qquad \sqrt{D^2 - D_i^2} = 9.68246 \text{ mm}$$

Bracket of the formula and denominator:

$$D - \sqrt{D^2 - D_i^2} = 10 - 9.68246 = 0.31754 \text{ mm} \qquad \pi D \left(D - \sqrt{D^2 - D_i^2} \right) = \pi \times 10 \times 0.31754 = 9.9759 \text{ mm}^2$$

Hardness:

$$HB = \frac{2P}{\pi D \left(D - \sqrt{D^2 - D_i^2} \right)} = \frac{2 \times 1000 \text{ kg}}{9.9759 \text{ mm}^2} = \mathbf{200.5 \text{ HB}} \approx 200 \; HB$$

The depth of the indentation is $h = 0.31754/2 = 0.1588$ mm and the area of the cap $S = \pi D h = 4.988$ mm², so $P/S = 1000/4.988 = 200.5$, the same.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $2P$ in kg (kgf) over an area in mm²: kgf/mm², the unit of hardness (no unit is attached to HB).
- **Load factor.** $Q = P/D^2 = 1000/10^2 = 10$, a value of the table of [Session 12 Slide 20] for $D = 10$ mm (load of 1000 kg), so the test is a standard one.
- **Limit.** For $D_i \ll D$, $D - \sqrt{D^2 - D_i^2} \approx D_i^2/(2D)$ and $HB \approx 4P/(\pi D_i^2) = 203.7$, close to the exact value (the cap is almost flat); for $D_i \to D$, the cap becomes a hemisphere, $S \to \pi D^2/2$, and $HB \to 2P/(\pi D^2)$.
- **Geometry.** $D_i/D = 0.25$: the indentation is a small fraction of the ball, consistent with a hard steel.

Interpretation

The steel has a Brinell hardness of about 200 kgf/mm². It is harder than the plain carbon steel of the table of typical values, 150 HBW [Session 12 Slide 28]. Hardness and tensile strength are both indicators of the plastic deformation, with a relation that depends on the material [Session 12 Slide 30].

Check against the official solution

Item	Official	Computed	Comment
Brinell hardness	200 HB	200.5 HB	agrees

`Problem 15` `Problems T5_MechanicalProperties.pdf #15` `Brinell hardness · inverse problem`

Indentation Diameter that Gives a Brinell Hardness of 300

`Original statement · Problems T5_MechanicalProperties.pdf, p. 5`

15. What will be the diameter of the indentation that produces a Brinell hardness of 300 when a load of 500 kg is applied? Diameter of the ball: 10 mm.

Official Solution: $d = 1.453$ mm

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. Same definition as in Problem 14: $HB = P/S$, with S the area of the spherical cap, $HB = \dfrac{2P}{\pi D\left(D - \sqrt{D^2 - D_i^2}\right)}$ [Session 12 Slide 19].
2. P in kilogram-force, lengths in mm; the hardness number has no unit attached.
3. Only the physically meaningful root is retained: the indentation diameter is positive and not larger than the ball, $0 < D_i \leq D$.

Data

Quantity	Symbol	Value
Brinell hardness	HB	300
Load	P	500 kg
Ball diameter	D	10 mm

Unknown: the diameter of the indentation D_i (called d in the key).

`PHASE 2 · FRAMES & STRATEGY`

This is the *inverse* of Problem 14: the hardness is known, and the indentation is the unknown. The formula of the slide is solved for D_i before substituting numbers:

1. Isolate the bracket: $D - \sqrt{D^2 - D_i^2} = \dfrac{2P}{\pi D\, HB} \equiv k$.
2. Isolate the square root: $\sqrt{D^2 - D_i^2} = D - k$.
3. Square both sides, $D^2 - D_i^2 = (D-k)^2$, and solve: $D_i^2 = D^2 - (D-k)^2 = 2Dk - k^2 = k\,(2D - k)$.

The squaring step is allowed here because $D - k > 0$ (the left side is a square root, and k is a small fraction of D), so no spurious root is introduced.

PHASE 3 · STEP-BY-STEP DERIVATION

1. The bracket k

$$k = D - \sqrt{D^2 - D_i^2} = \frac{2P}{\pi D\, HB} = \frac{2 \times 500}{\pi \times 10 \times 300} = \frac{1000}{9424.78} = 0.106103 \text{ mm}$$

2. The indentation diameter

$$D_i^2 = k\,(2D - k) = 0.106103 \times (20 - 0.106103) = 0.106103 \times 19.893897 = 2.11081 \text{ mm}^2$$

$$D_i = \sqrt{2.11081 \text{ mm}^2} = \mathbf{1.4529 \text{ mm}}$$

3. Verification by substitution

$\sqrt{D^2 - D_i^2} = \sqrt{100 - 2.11081} = 9.893897$ mm, which is $D - k = 10 - 0.106103 = 9.893897$ mm, as imposed in step 2. Putting $D_i = 1.453$ mm back in the formula:

$$HB = \frac{2 \times 500}{\pi \times 10 \times (10 - 9.893897)} = \frac{1000}{\pi \times 10 \times 0.106103} = 300.00$$

which recovers the 300 of the statement.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** k is a length (kg over kgf/mm² gives mm² and, divided by mm, a length); $D_i^2 = k(2D - k)$ is in mm².
- **Load factor.** $Q = P/D^2 = 500/10^2 = 5$, a standard value of [Session 12 Slide 20] for the ball of 10 mm.
- **Limit.** Since $k \ll D$ here, $D_i^2 \approx 2Dk = 4P/(\pi\, HB)$, so $D_i \approx \sqrt{4 \times 500/(\pi \times 300)} = 1.4567$ mm: the approximation is within 0.3 % of the exact value (1.4529 mm).
- **Direction.** A harder material (larger HB) leaves a smaller indentation under the same load: $D_i \propto HB^{-1/2}$ for small indentations.

Interpretation

A hardness of 300 HB under 500 kg leaves an indentation of only about 1.45 mm in a ball of 10 mm, with a depth of $h = k/2 = 0.05305$ mm. The ratio $D_i/D = 0.145$ is a very small indentation, which is difficult to measure accurately with a microscope (extension: the usual range of validity of the Brinell test is $0.25D$ to $0.6D$, a rule that is not in the slides).

Check against the official solution

Item	Official	Computed	Comment
indentation diameter	1.453 mm	1.4529 mm	agrees (the key rounds to three decimals)